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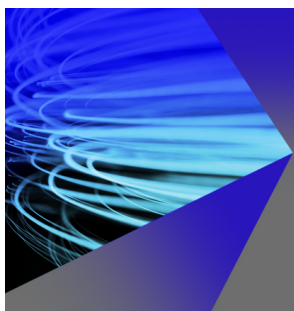
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ABSTRACT

The application of unmanned aerial vehicles (UAVs) has gained considerable momentum in recent years, with a marked reduction in costs, attracting growing interest from diverse fields. Nevertheless, as their prevalence continues to rise, amateur UAVs operating recklessly at low altitudes have become a serious threat to public safety. As a result, an economically efficient preventive solution is urgently required to identify potential UAV-related threats at an early stage. Acoustic-based detection offers a cost-effective and easily deployable solution on a large scale while also addressing certain limitations of other technologies (such as radar, radio frequency, and visual systems), thus holding considerable research significance. In this work, we review the research developments in acoustic-based UAV detection, highlighting the involved key technologies of UAV acoustic signals, such as signal acquisition, blind source separation, feature extraction and recognition, and sound source localization. A wide range of technical methods and application strategies employed in existing research are thoroughly covered, with an organized analysis of the relevant technological framework, detailing its core principles and representative methods, and comparing their merits and limitations in addressing UAV detection tasks. In addition, the challenges and problems faced by acoustic-based UAV detection are examined, with a discussion of the development paths required to fulfill its anticipated future potential.

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I. INTRODUCTION

Unmanned aerial vehicles (UAVs), colloquially referred to as drones, possess the capability of remote manipulation within predetermined boundaries,¹ with their genesis rooted in military applications.² The progression of multiple advanced technologies in recent years has facilitated the evolution of UAVs into more operationally simplified and economically viable systems, resulting in their extensive utilization and growing interest across civilian and commercial sectors. Presently, the UAV sector is witnessing unprecedented growth, with its utilization extending to various fields: agricultural management (crop spraying, farmland surveillance, and automated seeding),³ logistics and transportation (last-mile delivery and traffic monitoring),⁴ emergency services (disaster relief, fire suppression, and law enforcement),⁵ media production (news coverage, live broadcasting, and cinematic photography),⁶ and critical infrastructure maintenance (power grid inspection, pipeline integrity

assessment, and water resource monitoring).⁷ Despite their remarkable contributions to industrial productivity enhancement, UAVs have concurrently generated critical public safety risks, presenting formidable obstacles to effective oversight and preventive measures.

The exponential growth of UAVs in airspace over recent years has resulted in a surge of illegal drone activities, generating substantial apprehension.⁸ For example, the year 2019 witnessed substantial operational disturbances at airports across the United States, the United Kingdom, Ireland, and the United Arab Emirates due to unauthorized drone intrusions.⁹ A parallel event occurred in 2021 when the Iraqi prime minister's residence fell victim to a UAV-based assault. UAVs are highly susceptible to exploitation for various criminal activities, primarily including unauthorized entry into restricted airspace, privacy violations, contraband smuggling, terrorist attacks, and espionage operations.¹⁰ These developments constitute a critical hazard to community safety, carrying

profound impacts. Despite the implementation of comprehensive regulatory protocols by multiple countries to limit illegal UAV utilization, these initiatives still fall short of effectively countering intentional criminal conduct and terrorist operations. Consequently, regulatory authorities urgently require technologies capable of effectively detecting and tracking unauthorized UAVs,¹¹ which has significantly propelled the advancement of drone detection systems.

Current UAV detection methodologies primarily encompass radar tracking surveillance,¹² radio frequency (RF) monitoring,¹³ optical identification,¹⁴ and acoustic detection.¹⁵ Figure 1 provides a comprehensive summary of these technologies, each demonstrating distinct advantages and limitations. Although radar-based detection is commonly acknowledged as the most effective approach for remote, precise monitoring,¹⁶ its deployment entails significant costs and frequently lacks the capability to discriminate between amateur drones and ornithological entities. RF-based detection enables simultaneous localization of both UAVs and their operators,¹⁷ nevertheless, it is susceptible to interference from ground clutter in low-altitude environments and incapable of detecting autonomous drones.¹⁸ Optical-based detection renders visual recognition an exceptionally attractive methodology, offering capabilities in identification, tracking, and even payload inspection;¹⁹ however, it is highly susceptible to adverse weather conditions (e.g., dust, fog, and low light), obstructions, and UAV flight velocity.²⁰ In contrast, acoustic-based detection is characterized as a cost-effective, all-weather technology with relatively high precision. Relative to the aforementioned methodologies, acoustic technology demonstrates resilience against challenging conditions such as speed fluctuations, adverse weather patterns, and electromagnetic noise, successfully mitigating the deficiencies of other detection technologies. In particular, within integrated detection systems employing multiple technologies, acoustic-based detection effectively addresses the coverage gaps present in alternative detection methods. Significantly, acoustic-based detection technology, operating as a passive sensing method, maintains complete non-interference with concurrent

detection mechanisms and proximate communication infrastructures. Consequently, investigating and applying acoustic detection methodologies constitute an indispensable technical component in UAV regulation.

The acoustic-based UAV detection technology processes the acoustic signals captured by sensor arrays, distinguishing the distinctive features of UAV acoustic patterns from ambient noise,²¹ thereby enabling effective identification and spatial localization of UAVs. Generally, the methodology fundamentally manipulates audio signals and isolates discriminative features to optimize detection capabilities. Pioneering studies in this detection technology are predominantly centered around classical machine learning (ML) techniques, notably support vector machines (SVMs),²² K-nearest neighbors (KNNs),²³ multilayer perceptrons (MLPs),²⁴ and hidden Markov models (HMMs),²⁵ establishing fundamental frameworks for recognizing UAVs. Yet, such techniques exhibit substantial performance limitations when confronted with sophisticated noise conditions and the demands of large-scale dataset training. With the rise of deep learning (DL) techniques over recent years, substantial advancements have been achieved in UAV acoustic detection, where deep learning architectures, including convolutional neural networks (CNNs),²⁶ recurrent neural networks (RNNs),²⁷ and convolutional recurrent neural networks (CRNNs),²⁸ have exhibited exceptional complex pattern learning abilities, significantly boosting the system's noise immunity. In addition, research efforts have concentrated on blind source separation (BSS)²⁹ methodologies to address multi-UAV detection, effectively distinguishing between different drone signatures within mixed acoustic environments, which facilitates simultaneous target identification. Notably, incorporating sound source localization methodologies enables retrieving comprehensive spatial data³⁰ regarding UAVs, leveraging time-of-arrival differences, phase disparities, and angle-of-arrival information to trace UAV trajectories in both two- and three-dimensional space.

Capitalizing on the technological merits of acoustic-based detection, relevant methodologies have emerged as a progressively

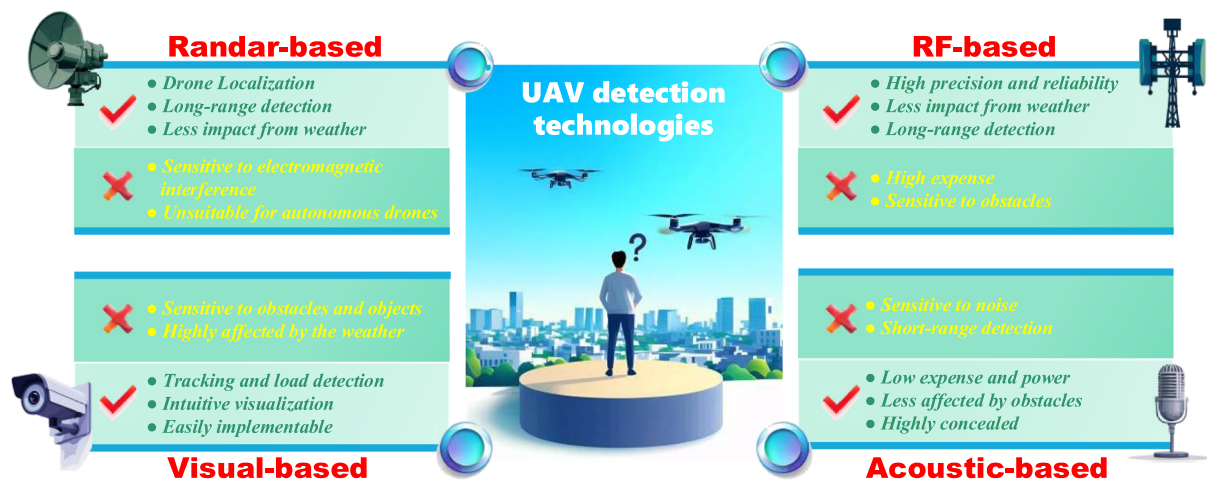


FIG. 1. Current technological framework for UAV detection.

vital component in drone detection systems, exhibiting broad potential for practical implementation. Extensive research has confirmed that acoustic-based detection approaches, through cross-domain integration with other technologies, offer enhanced UAV monitoring solutions. Employing multi-intelligent algorithms in UAV acoustic monitoring extends beyond mere presence identification, allowing for the recognition of specific drone types, flight trajectories, and behavioral assessment in controlled airspace. Deployment of lightweight, power-efficient acoustic array technologies facilitates large-scale implementation of acoustic-based UAV monitoring systems. Yet, as of now, in-depth summaries and investigations of acoustic-based UAV detection techniques remain scarce. The accelerated advancement of acoustic detection technology, coupled with the lack of systematic review and consolidation of numerous emerging theoretical approaches, has motivated us to initiate this detailed research endeavor. The key contributions of this work are outlined as follows:

- (1) A comprehensive review of the acoustic-based UAV detection technology framework is conducted, encompassing theoretical foundations, conceptual definitions, methodological approaches, and algorithmic equations.
- (2) Methodical categorization and comparison of diverse techniques in acoustic-based UAV detection are performed, spanning from signal acquisition and processing to feature extraction, deep learning architectures, blind source separation algorithms, and acoustic localization methods, covering both traditional approaches and emerging technologies.
- (3) An analysis of acoustic-based UAV detection applications is examined, accompanied by an exploration of prevailing challenges and promising future directions.

The overall layout of this work is depicted in Fig. 2. The remainder of this paper is organized as follows: Sec. II introduces the characteristics, acquisition methods, and datasets related to UAV acoustic signals. Section III consolidates UAV's acoustic signal processing methods and feature extraction technologies. Section IV provides a detailed discussion of existing UAV acoustic recognition and detection technologies from the perspective of signal feature learning. Section V delves into the theoretical foundation of sound source localization techniques and their utilization in acoustic-based UAV detection. Section VI presents future directions and their challenges for acoustic-based UAV detection. This paper concludes with Sec. VII, offering a comprehensive summary of the research.

II. UAV'S ACOUSTIC SIGNAL ACQUISITION

A. Acoustic characteristics of UAVs

UAV acoustic properties mainly consist of low-frequency and high-frequency signal components, exhibiting sound pressure levels between 60 and 90 dB, characterized by broadband and narrow-band spectral features, periodic modulation patterns, directional properties, and sensitivity to environmental conditions and aircraft configurations. These attributes are essential for drone detection, recognition, and system design. The dominant noise components in UAVs are generated by their rotating blades and electric motors.³¹ This is attributed to the fact that during aerial operation, the

high-speed rotation of UAV propellers creates low-frequency aerodynamic noise, whereas motor functioning emits high-frequency electromagnetic noise. The superposition of noise components at different frequencies results in UAV acoustic signals exhibiting distinct harmonic characteristics.³² The noise spectrum consists of line spectra and their interconnecting components, where harmonic frequencies appear as distinct spectral lines. These acoustic signatures often display broadband features with substantial spectral information,³³ a characteristic that enhances time-domain cross correlation localization techniques. Through appropriate signal processing approaches, mainly the Fourier transform and Mel-frequency cepstral coefficient analysis, UAV acoustic signatures are generally capable of being clearly differentiated from interference signals and ambient noise. Consequently, the unique characteristics of acoustic signals from UAVs make it possible to detect a range of multi-rotor and fixed-wing drones using acoustic signals and their corresponding processing methods. Furthermore, the UAV type, aircraft status, environmental conditions, and design materials are essential factors influencing its acoustic properties. These aspects offer a crucial foundation for detecting, localizing, and tracking multiple UAVs.

B. Acoustic sensors

Acoustic sensors, serving as crucial instruments for acquiring UAV acoustic signals, significantly influence algorithm selection, resource allocation,³⁴ and detection performance in monitoring tasks. The selection of appropriate sensors adheres to a series of criteria, principally involving sensitivity, frequency response (FR), signal-to-noise ratio (SNR), dynamic range, directivity, environmental durability, compatibility, size, shape, cost, and power consumption. In contemporary drone acoustic monitoring applications, sensors are typically classified according to their working mechanisms as capacitive,³⁵ piezoelectric,³⁶ fiber-optic,³⁷ and micro-electro-mechanical system (MEMS)³⁸ based acoustic detectors. Capacitive sensors operate by utilizing sound-induced vibrations on the eardrum, resulting in a change in the distance between the capacitor electrodes. This variation in polarization voltage is then used to convert the sound signal into an electrical signal. These sensors are characterized by high sensitivity, a broad frequency range, and low noise, with electret capacitive sensors being especially popular for various acoustic detection tasks due to their low cost. Piezoelectric acoustic sensors function by generating electrical signals from mechanical deformation. They are inexpensive, small in size, and require no external power, making them well-suited for long-term, unattended monitoring. However, their lower sensitivity and restricted high-frequency response (with significant signal attenuation above 10 kHz) limit their overall detection precision. Fiber-optic acoustic sensors rely on optical waveguide effects and acoustically induced light modulation for non-electrical sensing. Their primary benefits include long-range detection, high sensitivity, and strong resistance to interference, particularly in extreme electromagnetic environments such as those near substations or radar stations, making them highly suitable for drone surveillance. Although the cost of optical fibers is relatively low, the widespread civilian deployment of this technology is constrained by the expensive demodulation equipment required. MEMS is a micro-sized

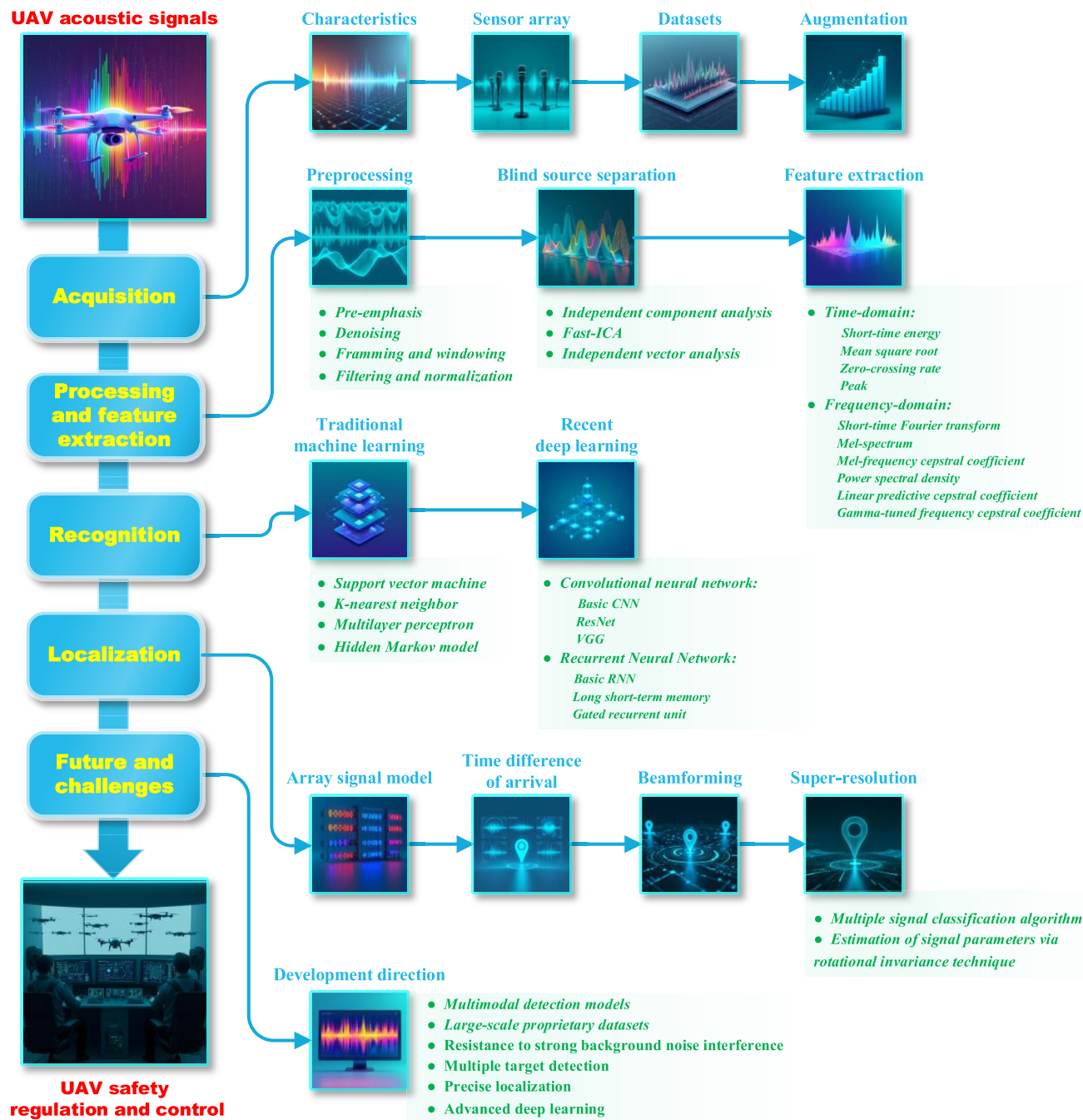


FIG. 2. Overall framework of this work.

acoustic sensor produced through semiconductor fabrication techniques. Its cost-effectiveness in mass production and ease of integration make it particularly well-suited for embedded mobile monitoring systems or integrated monitoring networks, holding a crucial and irreplaceable role in drone detection.

Table I provides a comparative analysis of acoustic sensors with different operational principles in UAV detection scenarios, evaluating their performance across five key parameters: sensitivity, frequency range (FR), signal-to-noise ratio (SNR), cost, and size. As can be observed, capacitive-based sensors possess excellent sensitivity

TABLE I. Performance comparison of prevalent acoustic sensors for UAV detection.

Category	Performance				
	Sensitivity	FR	SNR	Cost	Size
Capacitive	Moderate	Moderate	High	Low	Moderate
Piezoelectric	High	Wide	Moderate	Moderate	Small
Fiber-optical	Extremely high	Wide	Extremely high	High	Large
MEMS	Moderate	Wide	Moderate	Low or moderate	Extremely small

characteristics, although with significant sensitivity to environmental noise, thus being ideal for UAV detection in low-noise scenarios. Conversely, piezoelectric-based sensors demonstrate superior performance for UAV detection in high-noise scenarios. Optical fiber-based sensors are capable of achieving high-accuracy remote monitoring for UAVs, enabling consistent data collection for prolonged durations. MEMS-based sensors exhibit distinctive miniaturization properties, rendering them especially appropriate for portable detection equipment and onboard drone surveillance applications.

C. Acoustic signal datasets of UAVs

UAV acoustic signal datasets typically encompass sound signatures generated during various flight states (e.g., hovering, ascending, descending, and level flight), collected through high-sensitivity microphone arrays, covering multiple drone models, flight altitudes, velocities, and environmental conditions (including indoor, outdoor, and scenarios with or without background noise). Commonly accessible, relevant datasets are primarily derived from publicly available databases developed through dedicated research efforts. As presented in Table II, these datasets, although limited in quantity, contain substantial UAV acoustic recordings accompanied by detailed labeling information (including drone types, flight states, timestamps, etc.), facilitating research and algorithm development for UAV detection, classification, and trajectory monitoring. For

instance, according to references,³⁹ a comprehensive publicly available dataset has been developed, allowing for the classification of various UAV audio signals and environmental sounds in multi-UAV scenarios, thus contributing to system-level decision-making and autonomy. The dataset in Ref. 40 employs advanced Generative Adversarial Networks (GANs) to generate large volumes of UAV audio data, providing a new solution to the challenge of data scarcity. The dataset mentioned in Ref. 41 offers a multimodal data collection that integrates both video and audio, enhancing detection reliability in complex environments. Reference 42 introduces a public dataset for drone sound source localization, filling the gap in existing datasets in terms of standardized sound source localization.

In addition, some databases encompass sound signatures from different drone categories, such as multi-rotor and fixed-wing aircraft, combined with simulated environmental noise and disturbance signals in authentic settings, offering comprehensive data resources for scientific investigations.

D. Data augmentation

Current drone acoustic detection research faces challenges such as limited annotated data, uneven distribution, and scenario homogeneity. These constrained datasets hinder the deployment of learning algorithms and advanced classifiers that rely on large training datasets. Data augmentation offers a practical strategy to overcome

TABLE II. Overview of diverse UAV acoustic signal datasets.

Data source	Sampling frequency	Datasets information	Note	References
Self-built open-access	44.1 kHz (16-bit)	7 types of multi-rotor UAVs and various noises	Use manually mixed audio	15
Self-built open-access	None	44 drone audio samples and different environmental sounds	4924 UAV sound files and 23.42 total playtime	39
Self-built open-access	44.1 kHz (64-bit)	Collected Bebop and Mambo drone audio and ESC ambient audio	1300 drone audio files and 2000 environmental audio files	40
Self-built open-access	None	Capture drone recordings from different distances	Drone audio every 5 m (5–50 m)	43
Self-built open-access	44.1 kHz	Collecting datasets for multi-sensor drone inspection	Contains 650 videos and 90 audios	41
Self-built open-access	44.1 kHz	17 UAV variants	Each drone flies at an altitude of 8 m	44
Self-built	96 kHz (32-bit)	Drone audio for sound source localization	The audio duration is 33 090 s, totaling 31.95 GB.	42

this limitation by generating, modifying, synthesizing, or transforming existing data to increase dataset volume and heterogeneity, thereby providing data support to enhance the generalization capability and robustness of machine learning systems. For instance, Kümritz⁴⁵ significantly expanded the quantity and quality of original data by introducing techniques such as harmonic distortion, pitch shifting, temporal delay, and background noise mixing.

Building upon traditional data augmentation methods, deep learning-based generative models offer more powerful tools to address drone acoustic data scarcity. Generative Adversarial Network (GAN), Variational Autoencoder (VAE), and their variants are among these tools, while diffusion models—which have gained prominence in recent years—also play a key role. Collectively, these models fundamentally resolve data scarcity by learning the intrinsic distribution of raw data, enabling them to synthesize samples that are in feature space and indistinguishable from real data. GANs continuously refine the generator's capabilities through adversarial mechanisms, while VAEs, based on probabilistic graphical models, ensure sample diversity by learning the latent space of data and performing interpolation or sampling. Diffusion models, meanwhile, generate samples through an incremental denoising process and currently rank among the highest fidelity models in audio and image generation. As a result, researchers have already applied these generative models to augment drone audio data. For instance, Al-Emadi *et al.*⁴⁶ employed GANs to generate realistic drone audio datasets, thereby enhancing the diversity of original datasets and significantly improving model classification performance.

III. ACOUSTIC SIGNAL PROCESSING AND FEATURE EXTRACTION FOR UAVS

The raw acoustic signal data captured from UAVs usually exhibit high complexity, often containing distortions and aliasing phenomena caused by data collection processes, which markedly elevate the challenges in subsequent drone categorization. Therefore, substantial research efforts in acoustic-based UAV detection have concentrated on signal processing and feature extraction strategies, targeting the accurate identification of critical information related to drone detection within complex acoustic patterns, consequently fostering the emergence of various signal processing techniques and their field applications. This section explores how to transform complex raw acoustic signals into efficient, learnable features through three key aspects: preprocessing, blind source separation, and feature extraction. It distills the rationale and steps of preprocessing to enhance the audio signal-to-noise ratio and separability. Subsequently, it introduces BSS modeling and representative algorithms to recover target sound sources from complex reverberation and multi-source interference. Finally, it compares the discriminative power, noise robustness, and computational overhead of commonly used time-domain and frequency-domain features, providing a robust signal processing framework and feature engineering foundation for subsequent drone acoustic classifier design.

A. Preprocessing

Within the realm of UAV acoustic detection, the preprocessing of signals represents a fundamental step, principally dedicated to extracting relevant features from unprocessed audio inputs and

reducing noise interference to augment the accuracy of subsequent identification and classification tasks. The typical preprocessing workflow involves pre-emphasis, noise suppression, framing and windowing, filtering, normalization, and time-frequency domain representation,⁴⁷ intended to eradicate environmental noise and intensify target signal attributes, thus converting the signals into more appropriate formats for analytical purposes. By implementing preprocessing, the differentiation capability of drone acoustic patterns is markedly enhanced, yielding high-quality input for ensuing signal transformation, feature extraction, and machine learning model development, thereby improving detection capabilities and system robustness.

1. Pre-emphasis

This technique enhances high-frequency components within acoustic signals, generally achieved through a first-order high-pass filtering mechanism. For UAV acoustic monitoring, the originally acquired signals exhibit energy concentration primarily in low-frequency bands, with limited high-frequency representation. Pre-emphasis application⁴⁸ partially mitigates high-frequency signal loss, improving the quality of acoustic characteristic parameters.

2. Denoising

Target sound events in acoustic environments are often masked by ambient noise, resulting in compromised drone detection capabilities, a critical issue that must be addressed. Denoising plays an important role in drone acoustic detection, as it ensures that purer, more distinguishable features are extracted, thereby improving the performance of subsequent classification or recognition models. For example, for purer UAV acoustic signal acquisition, multiple studies⁴⁹ have implemented BSS methods to segregate mixed acoustic inputs, successfully eliminating extraneous noise. Furthermore, a study⁵⁰ has proposed a methodology based on the linear shrinkage-subspace projection-power spectral density filter (LSP) that employs covariance matrices and enhanced Wiener filtering, achieving effective noise suppression within a 135-m radius.

3. Framing and windowing

In real-world scenarios, acoustic signals often exhibit non-stationary behavior due to the dynamic changes in their statistical properties over time. As a result, the signal is typically divided into small, short-time frames to ensure approximate stationarity within each frame and facilitate better feature extraction. The boundary effects caused by segmentation, also known as truncation effects, are mitigated by applying window functions to the frames. This helps to smooth the signal edges, reduce spectral leakage, and improve the accuracy of frequency-domain analysis. Frequently used window functions include the rectangular, Hamming, Hanning, and Blackman windows.⁵¹

4. Filtering and normalization

Filtering and normalization are crucial preprocessing steps in UAV acoustic detection, aimed at improving the quality of audio features from the perspectives of frequency-domain denoising and time-domain amplitude normalization. The recorded UAV noise signals often contain substantial environmental noise and interference, which can disrupt feature extraction and recognition, thus reducing detection accuracy. On one hand, filtering techniques can

design low-pass, high-pass, and bandpass filters based on different frequency response ranges, selectively preserving or suppressing signals from specific frequency bands to eliminate noise interference. To effectively cope with noise interference and non-stationary characteristics in complex signal environments, nonlinear system modeling is used, dynamically adjusting parameters for adaptive filtering.⁵² UAV noise signals also change significantly with varying distances from the UAV, with changes in signal strength causing noticeable feature variations. This leads to issues such as signal attenuation and feature distortion, significantly increasing the training data requirements. On the other hand, normalization focuses on adjusting the amplitude or energy distribution of signals to eliminate discrepancies in the numerical range of audio information. In UAV detection, this can effectively mitigate the impact of distance and enhance feature stability,⁵³ including methods such as peak normalization, feature-level standardization, and dynamic response normalization.

B. Blind source separation (BSS)

Robust signal feature recognition in UAV acoustic detection faces substantial obstacles in field deployments because of intricate background noise interference patterns. Against this backdrop, the blind source separation (BSS) technique emerges as an innovative solution for enhancing the performance of acoustic-based UAV detection. This technique aims to recover constituent source signals from observed mixtures while not requiring prior knowledge of the mixing process.⁵⁴ Such a framework consequently permits autonomous separation of UAV-specific acoustic patterns from complex noise backgrounds without environmental priors, substantially boosting the detection system's signal feature identification reliability. It demonstrates sufficient resolution to discriminate between different drone sound profiles, facilitating simultaneous monitoring of multiple UAVs.

1. BSS fundamental principle

The fundamental objective of BSS is to recover the original source signals from observed mixtures without requiring prior knowledge of either the mixing process or the source characteristics.⁵⁵ BSS research commenced with Cherry's seminal 1953 cocktail party problem,⁵⁶ illustrating the human auditory system's remarkable capacity to segregate concurrent speech streams amid acoustic clutter. The architectural framework for applying BSS to acoustic-based UAV detection is visually summarized in Fig. 3.

For the mathematical modeling of BSS, we define m as the undetermined quantity of acoustic source signals (S), n as the number of acoustic sensor recordings (X), and t as the time series, represented by

$$\begin{cases} S = [s_1(t), s_2(t), \dots, s_m(t)]^T, \\ X = [x_1(t), x_2(t), \dots, x_n(t)]^T. \end{cases} \quad (1)$$

Based on the above specifications, the observed mixtures $x_i(t)$ and source signals $s_i(t)$ adhere to the following basic BSS formulation:

$$\begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \cdot \begin{bmatrix} s_1(t) \\ s_2(t) \\ \vdots \\ s_n(t) \end{bmatrix} + \begin{bmatrix} b_1(t) \\ b_2(t) \\ \vdots \\ b_n(t) \end{bmatrix}, \quad (2)$$

where a denotes the mixing matrix element encoding the complete set of coefficients determining the source signal combination, and b represents the optional additive noise component.

For convenience, the above formula is simplified to

$$X = A \cdot S + B, \quad (3)$$

where A and B refer to the mixing matrix and the noise matrix, respectively.

The term "blind" in BSS implies no prior knowledge of source signals and mixing matrix, with the core objective being to estimate these elements solely from observed sensor signals.⁵⁷ BSS models can be categorized into three distinct classes based on the relationship between observed signals (n) and sources (m): overdetermined ($n > m$), exactly determined ($n = m$), and underdetermined ($n < m$) scenarios. The aforementioned derivations are established under exactly determined conditions. For overdetermined scenarios, dimensionality reduction techniques such as whitening based on principal component analysis (PCA) or non-negative matrix factorization (NMF) are typically employed to achieve the required exactly determined configuration. For underdetermined scenarios, conventional approaches employ regularization techniques, sparse representation methods, and composite signal construction (observable signal synthesis) to address the inherent mathematical challenges.

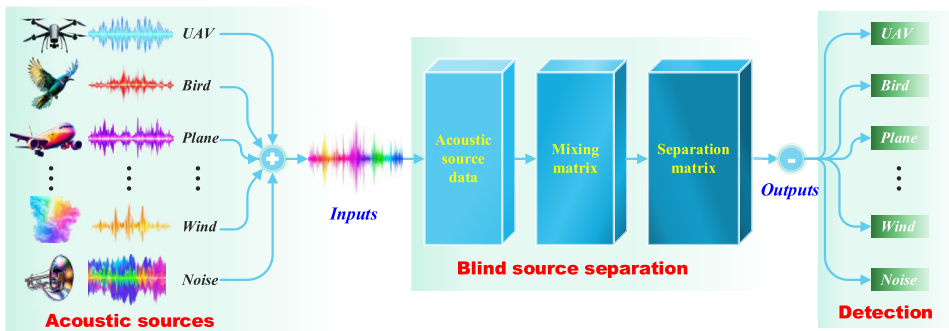


FIG. 3. BSS flow chart in acoustic-based UAV detection.

2. Independent component analysis (ICA)

As one of the key technologies for blind source separation, independent component analysis (ICA) has found widespread applications in fields such as audio signal processing, medical signal processing, and image processing. Recently, it has shown promising results in drone acoustic detection, providing effective source separation and noise suppression, significantly enhancing the interference resistance of detection systems. Its application in this area is gaining increasing attention. The separation mechanism of ICA presupposes complete statistical independence among source signals,⁵⁸ excluding all forms of cross correlation. Through appropriate mixing matrix formulation and objective function maximization, the algorithm achieves blind recovery of constituent signals through unsupervised optimization. Statistical independence serves as ICA's fundamental postulate. Since independent signals exhibit zero statistical correlation, the central limit theorem dictates that their linear combinations typically demonstrate more excellent Gaussian characteristics than any individual source signal. Consequently, extracting signals with more substantial non-Gaussian characteristics facilitates the identification of independent source components.⁵⁹ Common non-Gaussianity metrics include entropy, kurtosis, and skewness for quantitative evaluation.

The ICA aims to retrieve the source signals S from the observed mixtures X , which requires estimating a separation matrix W . The relationship between these components can be understood through the following equation:

$$Y = W \cdot X = W \cdot A \cdot S = C \cdot S, \quad (4)$$

where $Y = [y_1(t), y_2(t), \dots, y_n(t)]$ denotes the estimated source signal vector. The product of matrices W and A is represented as a unit matrix C under ideal conditions.

In addition, ICA works by maximizing the non-Gaussianity of the signals to estimate the separation matrix. In general, sharper kurtosis and higher negative entropy correspond to stronger non-Gaussianity of the signals. The non-Gaussian measure for ICA, formulated using negative entropy, is given as follows:

$$J(s) = H(y_{\text{Gauss}}) - H(s), \quad (5)$$

where $J(s)$ represents the source signals' negative entropy, H indicates the entropy, and y_{Gauss} refers to a Gaussian random variable.

The iterative update formula for ICA is commonly expressed as

$$w_{k+1} = w_k + \eta \left\{ E \left[xg \left(w_k^T x \right) \right] - E \left[xg' \left(w_k^T x \right) \right] w_k \right\}, \quad (6)$$

where w is the separation vector in the separation matrix W ; the observed signal x is projected onto this direction, such that the projected signal y maximizes non-Gaussianity and independence; $g(x)$ is a nonlinear function, typically chosen based on the signal characteristics; and η is the learning rate.

ICA provides an effective speech signal separation approach, leading to its growing importance in acoustic-based UAV detection. Ghani and Khan⁶⁰ modeled mixing non-speech, non-Gaussian, and independent sources. Building on ICA, they adapted the projection pursuit algorithm to the acoustic detection domain, revealing significant advantages in drone acoustic features and successfully

recovering drone signals from mixed audio. This provides new strategies for drone monitoring in challenging acoustic environments. In another study, Uddin *et al.*⁶¹ proposed a time-varying UAV detection technique based on ICA. By combining ICA with a channel tracking mechanism, they effectively solved the signal separation problem in time-varying scenarios, achieved the detection of multiple UAVs, and improved the detection accuracy and adaptability of the system. Although ICA can separate simple noise mixtures, when assuming Gaussian-distributed background noise, the actual detection rate may drop significantly, positioning ICA as a more theoretical or simple-scenario solution.

3. Fast-ICA

Fast-ICA, a variant of ICA, offers a more efficient implementation of the ICA algorithm. While both ICA and Fast-ICA are grounded in the same theoretical framework, they adopt different optimization strategies. ICA uses gradient ascent iteration, while Fast-ICA optimizes using a fixed-point method. Negative entropy remains the non-Gaussian measure, and Fast-ICA's iterative update formula is as follows:

$$w_{k+1} = E \left[xg \left(w_k^T x \right) \right] - E \left[xg' \left(w_k^T x \right) \right] w_k. \quad (7)$$

Compared with ICA, Fast-ICA has a more streamlined algorithmic structure, eliminating the need for a learning rate, with the update step size being automatically determined. This overcomes the computational complexity and slow convergence rate found in ICA. Therefore, Fast-ICA is more adaptable to high-dimensional data or more complex models, making it particularly effective for UAV detection in complex acoustic environments.

Fast-ICA, known for its efficient computation, has become a widely adopted technique for separating various signal components in signal detection. Uddin *et al.*⁶² proposed an efficient machine learning method that combines Fast-ICA with a support vector machine (SVM) for spatial-domain acoustic-based UAV detection. After unmixing the audio signals, this method can detect various environmental sound signals and is also effective in detecting multiple UAVs, highlighting the potential of combining blind source separation with machine learning in UAV detection. Wang *et al.*²⁹ introduced a UAV acoustic detection method based on a blind source separation framework, with Fast-ICA at its core, and combined it with principal component analysis (PCA) and variational mode decomposition (VMD) techniques. This method addresses the challenges of underdetermined, determined, and overdetermined situations in acoustic-based UAV detection, offering a more comprehensive solution.

4. Independent vector analysis (IVA)

The independent vector analysis (IVA) method is an extension of ICA, defined as a tool for multivariate data processing. It expands the ICA framework to multi-dimensional datasets and is specifically designed to handle multi-channel signals. This method overcomes the traditional limitations of ICA and can effectively separate Gaussian or correlated sources. IVA assumes that the source signals in each channel are independent, although there may be statistical dependencies between different channels. By promoting the

time step to an l -dimensional form, the ICA model can be expanded to accommodate this higher-dimensional structure,

$$Y^l = W^l \cdot X^l, \quad (8)$$

where $l = [1, 2, \dots, L]$ represents the temporal dimension. Alternatively, the time step can be adjusted to represent different acquisition channels or other indices, such as $[Mic_1, Mic_2, \dots, Mic_L]$, where “Mic” refers to the sensor index in an acoustic sensor array. This approach allows for integrating data from multiple sensors, leveraging inter-channel statistical information and localization techniques to accurately detect the position and even the movement trajectory of UAVs.

IVA represents a relatively novel and promising approach to BSS, although notable research achievements in its application to acoustic-based UAV detection remain scarce. However, some progress in this area is still noteworthy. For example, Uddin *et al.*⁶³ introduced an IVA-based method for acoustic-based UAV detection, designed to unmix signals recorded from multiple microphones. It enables simultaneous detection of several UAVs under heavy interference and has shown better performance than ICA in experimental evaluations. The main limitation, however, lies in its high computational cost. To tackle the bottlenecks of environmental robustness and processing speed, Kang *et al.*⁶⁴ proposed a unified framework where IVA functions as the spatial model and deep neural networks serve as the source model, achieving real-time performance in mixed speech separation tasks. Scheibler⁶⁵ developed an enhanced auxiliary function-based IVA (AuxIVA) approach, incorporating a calibrated iterative projection algorithm (IPA), to accelerate convergence in blind speech separation under reverberant conditions. To improve the efficiency of speech signal separation, Brendel and Kellermann⁶⁶ proposed an accelerated IVA method that utilizes the majorize-minimize (MM) framework combined with

eigenvalue decomposition. IVA has proven to be highly effective in signal separation tasks. As its performance continues to advance, it is likely to become a foundational technology for complex multi-UAV detection scenarios in future airspace monitoring systems.

C. Feature extraction

Feature extraction refers to the process of extracting features from acoustic signals, aiming to integrate prior knowledge of acoustics, sound perception, or specific attributes into the audio scene. This process results in a compact data representation that enables efficient sound scene analysis. Broadly speaking, as depicted in Fig. 4, UAV acoustic features can be classified into two categories: time-domain features and frequency-domain features.

1. Time-domain features

Time-frequency features are those extracted from the amplitude variations of an acoustic signal as it changes over time. These features capture both the instantaneous fluctuations in the signal and its statistical characteristics over short time intervals. In addition, these features are straightforward to compute and analyze and are extensively used in UAV audio analysis. Some of the most common time-domain features include

a. *Short-time energy (STE)*. Energy represents a quantitative measure of the time-varying amplitude characteristics of a speech signal. It quantifies the overall strength and intensity of the signal, offering insights into amplitude fluctuations and contributing to a comprehensive understanding of the audio signal and its features. Short-time energy⁶⁷ is applied to short-time frames of the signal. It is beneficial for analyzing the dynamic changes of acoustic signals over time, especially for non-stationary signals such as UAV sounds. It is calculated by summing the squared values of the signal within a defined short-time window.

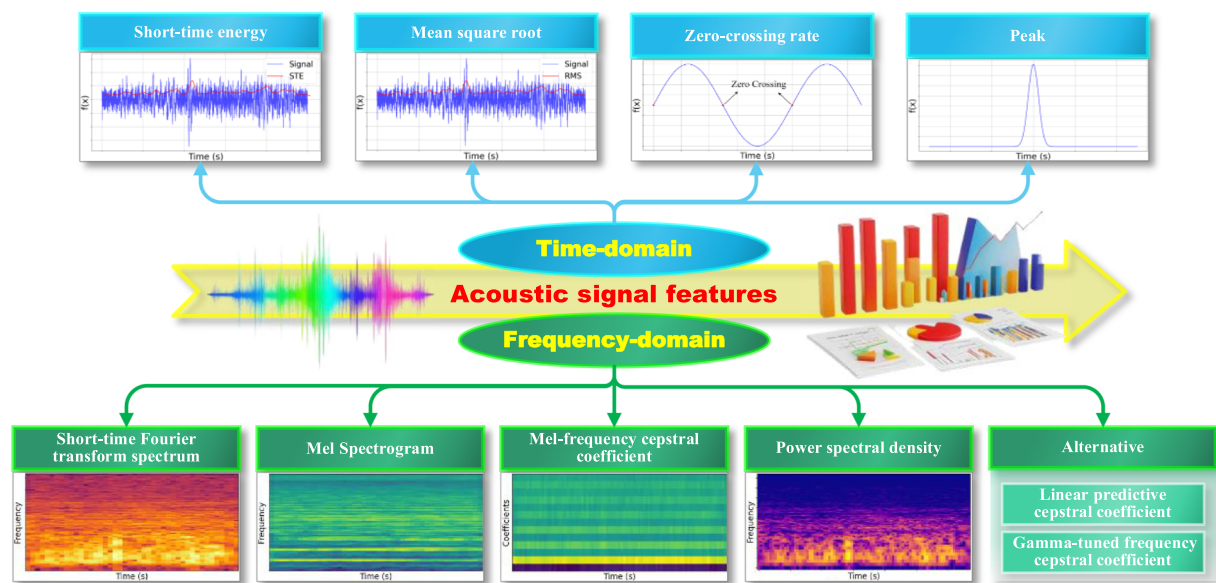


FIG. 4. Categorization of acoustic features for UAVs.

b. Root mean square (RMS). RMS serves as a fundamental indicator for assessing the average power of an audio signal, capturing the energy distribution across the time domain.⁶⁸ It is efficient in computing and effective in reducing the impact of non-periodic noise via squaring. However, it lacks the ability to differentiate harmonic signatures from UAV propellers and must be complemented with other features, such as spectral centroid, to enhance the classifier's generalization.

c. Zero-crossing rate (ZCR). ZCR measures the frequency at which a speech signal crosses the zero axis within a defined time period.⁶⁹ It is determined by counting the number of polarity changes in the signal within a particular window. This feature is useful for distinguishing between different phonemes and is simple to compute, making it effective for UAV audio recognition tasks. However, its performance can be negatively affected by noise, resulting in relatively poor robustness.

d. Peak. The peak reflects the maximum amplitude of an audio signal, representing the highest value observed within an audio segment.⁷⁰ It provides insight into the signal's intensity and dynamic range and is crucial in audio processing to prevent distortion resulting from overly high signal levels.

2. Frequency-domain features

Frequency-domain features are derived from signals represented in the frequency domain, specifically from their spectral characteristics. Spectral representations, being two-dimensional visual displays, illustrate the strength distribution of various frequency components in signals, offering direct insight into the principal frequency features of acoustic waveforms. Frequency-domain features provide comprehensive spectral information that accurately reflects the acoustic characteristics of UAVs, leading to their widespread application. Frequently observed features consist of the following:

a. Short-time Fourier transform (STFT) spectrum. The STFT spectrum⁷¹ is a local Fourier transform technique that segments the non-stationary acoustic signals of drones into frames, applies windowing, and then performs Fourier transforms. This process maps the signal into a time-frequency matrix, enabling the capture of dynamic features in both temporal and spectral domains. It is particularly effective for identifying transient characteristics such as sudden changes in propeller speed and harmonic components of motor noise. Furthermore, the STFT spectrum provides foundational data for subsequent feature extraction processes. The mathematical expression of STFT is given as follows:

$$STFT(t, f) = \int_{-\infty}^{+\infty} x(\tau)h(\tau - t)e^{-j2\pi f\tau}d\tau, \quad (9)$$

where t stands for the time variable, f represents the frequency variable, $x(\tau)$ is the original signal, and $h(\tau - t)$ refers to the applied window function.

b. Mel-spectrum. Mel-spectrum⁷² constitutes a short-time power spectral representation of acoustic waveforms. The transformation is accomplished by converting power spectrum frequencies to the Mel scale, intending to precisely emulate the human ear's nonlinear frequency perception characteristics, which demonstrates significant utility in speech processing and acoustic analysis

applications. For Mel-spectrum calculation, acoustic signals are generally divided into overlapping short-time frames, allowing the implementation of STFT for power spectral density estimation. The power spectrum is then processed through a Mel-scale filter bank, mapping frequency-specific power values to their respective Mel-band representations. The process concludes with calculating logarithmic Mel-scale power values, producing the final Mel-frequency spectrogram. Figure 5 illustrates the workflow of Mel spectrogram generation. Mathematical representations of the Mel scale and Mel filter bank are represented as follows:

$$f_{\text{Mel}} = 2595 \times \log \left(1 + \frac{f}{700} \right), \quad (10)$$

$$H_m(k) = \begin{cases} 0, & k < f(m-1), \\ \frac{k - f(m-1)}{f(m) - f(m-1)}, & f(m-1) \leq k \leq f(m), \\ \frac{f(m+1) - k}{f(m+1) - f(m)}, & f(m) < k < f(m+1), \\ 0, & k > f(m+1), \end{cases} \quad (11)$$

where f_{Mel} indicates the Mel frequency on the Mel scale; f denotes the linear frequency (in Hz); m signifies the filter identifier, satisfying the inequality $1 \leq m \leq M_{\text{Mel}}$, in which M_{Mel} represents the total count of filters within the Mel-scale filter bank; and $f(m)$ refers to the central frequency of the m -th Mel-scale filter. By implementing Mel-spectrum analysis, UAV acoustic signals are mapped to a perceptually significant domain, effectively boosting the performance of detection methodologies.

c. Mel-frequency cepstral coefficient (MFCC). As a prevalent audio feature representation, Mel-frequency cepstral coefficients (MFCCs) consist of 12 frequency-related cepstral coefficients,⁷³ offering an optimal balance between feature discriminability and representation compactness through adequate spectral channel information. MFCCs are obtained by applying a linear cosine transform (DCT) to the logarithmically scaled power spectrum on a nonlinear Mel-frequency scale, with the resulting coefficients collectively forming the Mel-frequency cepstrum. Key steps in MFCC feature extraction include analog-to-digital conversion, pre-emphasis processing, frame windowing, Fourier transformation, Mel-scale filtering, and discrete cosine transformation. By executing these procedures, MFCC features effectively capture essential spectral characteristics of speech signals while preserving relevant acoustic properties and eliminating redundancy.

d. Power spectral density (PSD). As a fundamental metric in signal analysis, power spectral density (PSD) quantifies the frequency-domain power allocation of signals,⁷⁴ elucidating the energy distribution pattern across spectral components. Serving as a pivotal characteristic in UAV acoustic detection, PSD proves highly effective in capturing low-frequency range propeller noise and associated harmonic features. This feature enables the identification of dominant frequency components in UAV acoustic signatures, facilitating both drone detection and classification.

e. Alternative features. Existing research investigations have utilized linear predictive cepstral coefficients (LPCCs),⁷⁵ gamma-tone frequency cepstral coefficients (GFCCs),⁷⁶ spectral envelope

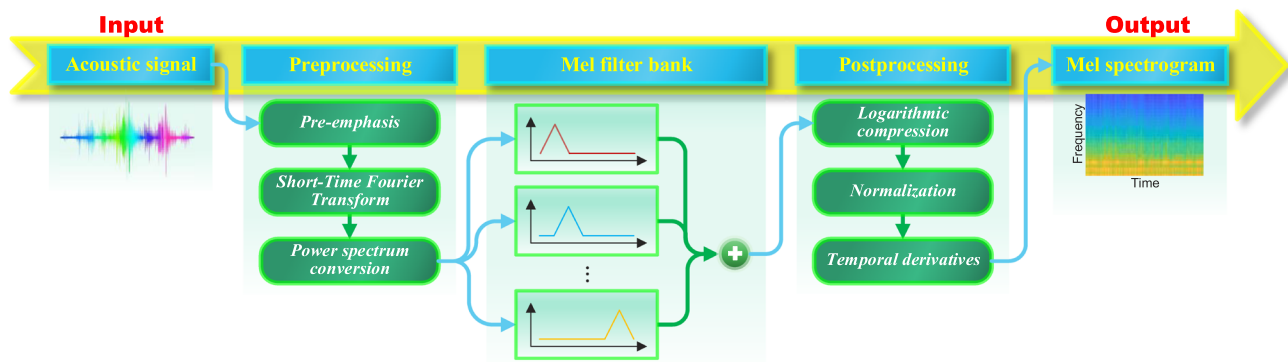


FIG. 5. Mel spectrogram workflow.

features,⁴² and wavelet transform coefficients⁷⁷ for UAV acoustic feature representation, exhibiting considerable efficacy in multiple operational scenarios.

In the context of UAV acoustic detection, frequency-based features demonstrate distinct advantages over time-domain features in identifying the signal's periodic patterns and harmonic components, accentuating dominant frequencies and providing greater efficacy for UAV sound analysis. Yet, the computational demands of frequency-domain processing often exceed those of time-frequency techniques, compromising the representation of instantaneous signal changes and temporal localization. The synergistic integration of both feature domains is anticipated to offer optimized solutions for acoustic-based UAV detection applications.

IV. UAV'S ACOUSTIC SIGNAL FEATURE RECOGNITION

The unique acoustic signal produced by the rotating blades of a UAV exhibits a broad frequency spectrum and abundant harmonics,⁷⁸ offering a distinct level of recognizability. Utilizing either traditional machine learning (ML) models or deep learning (DL) models to analyze features extracted from acoustic signals allows for the effective differentiation of UAV sounds from background noise or interference, leading to accurate UAV recognition. Below are some of the representative methodology frameworks.

A. Traditional machine learning-based methods

UAV recognition using traditional ML models for acoustic signal analysis has been demonstrated to be both efficient and accurate. Recently, conventional audio classification models such as support vector machine (SVM), K-nearest neighbor (KNN), multilayer perceptron (MLP), and hidden Markov model (HMM) have been increasingly used for UAV acoustic recognition, showing outstanding performance in various tasks.⁷⁹

1. Support vector machine (SVM)

SVM is a supervised learning algorithm designed for classification and regression tasks.⁸⁰ The core idea is to find an optimal hyperplane that maximizes the margin between different classes in the feature space. For linearly separable data, this involves solving a convex optimization problem to identify the hyperplane with the

largest possible separation from the nearest data points of each class, called support vectors. In cases where data are not linearly separable, SVM employs the kernel trick (e.g., polynomial or radial basis function kernels) to implicitly map inputs into a higher-dimensional space where separation becomes feasible. SVM is robust to overfitting, especially in high-dimensional spaces, and excels in tasks such as text classification and image recognition by focusing on the most informative training examples (support vectors) while disregarding outliers. It has been extensively applied in pattern recognition problems, such as acoustic-based UAV detection. For instance, Yang *et al.*⁸¹ designed a multi-node acoustic UAV detection system that uses SVM to recognize STFT features of acoustic signals, accomplishing complete drone detection coverage with no blind spots within a 75-m range. Shi *et al.*⁸² designed an acoustics-based UAV monitoring system, applying SVM to classify acoustic signals from a single microphone. The system achieved a maximum detection error of 6.45% and real-time UAV localization within 100 m, demonstrating the advantages of SVM's rapid classification. However, SVM is highly sensitive to feature selection and kernel function parameters, and the computational cost increases significantly when processing large-scale datasets, which limits its application in scenarios with high automation requirements.

2. K-nearest neighbor (KNN)

KNN is a simple yet powerful instance-based learning algorithm used for both classification and regression tasks.⁸³ The core principle is that similar data points tend to have similar outcomes. Given a new input, KNN identifies the K closest training examples (neighbors) in the feature space, typically using distance metrics such as Euclidean or Manhattan distance. For classification, the output is determined by majority voting among the K neighbors, while for regression, it is the average of their target values. KNN makes no assumptions about data distribution and is lazy learning (no explicit training phase), but its performance heavily depends on the choice of K, distance metric, and feature scaling. It excels in low-dimensional data but suffers from the curse of dimensionality as the feature space grows. It boasts advantages such as simplicity of principle, strong real-time update capability, and high scalability, demonstrating broad application prospects in acoustic-based UAV detection. For example, Akbal *et al.*⁸⁴ propose an acoustics-based

UAV classification model, utilizing Iterative Domain Component Analysis (INCA) to select features with higher informational content, followed by using KNN to achieve a classification accuracy of 99.72%. In addition, Uddin *et al.*⁶² introduce an efficient detection method based on ICA, which demixes the recorded audio to extract pure UAV sounds and employs KNN as a classifier, achieving a 99.1% accuracy in multi-UAV detection. Since KNN needs to traverse all training samples during each prediction, the computational overhead during the prediction phase is high, and an increase in feature dimensions severely affects the efficiency of distance calculation, known as the “dimension disaster,” which significantly affects its detection accuracy.

3. Multilayer perceptron (MLP)

MLP is a class of feedforward artificial neural networks composed of multiple layers of interconnected neurons.⁸⁵ Its core principle involves hierarchical feature transformation: input data are processed through one or more hidden layers, where each neuron applies a weighted sum of its inputs, followed by a nonlinear activation function (e.g., ReLU or sigmoid) to introduce complexity and enable the network to learn nonlinear relationships. The final output layer produces predictions, typically using a softmax (for classification) or linear activation (for regression). Training relies on backpropagation, which adjusts weights via gradient descent to minimize a loss function by propagating errors backward through the network. MLPs serve as universal function approximators, capable of modeling intricate patterns in tasks such as image recognition, regression, and time-series forecasting. Despite challenges in depth and efficiency, its flexibility and generality still make it of significant value in acoustics-based UAV recognition tasks. For example, Tejera-Berengue *et al.*⁸⁶ studied the impact of distance and environment on acoustics-based UAV detection, analyzing the recognition capabilities of five machine learning methods, including MLP, linear discriminant analysis, radial basis function network, SVM, and random forest, on UAV acoustic signals. The study confirmed the feasibility of using MLP for UAV recognition within a range of 200 m. Ahmed *et al.*⁸⁷ proposed a low-cost, highly reliable UAV detection system based on machine learning and, in a dataset containing seven different UAV sounds and environmental signals, demonstrated that MLP can effectively recognize the complex nonlinear acoustic features of UAVs. However, the training process of MLP is prone to getting stuck in local optima. It is susceptible to the selection of hyperparameters such as initial weights and learning rates, requiring extensive tuning work.

4. Hidden Markov model (HMM)

HMM is a probabilistic graphical model used for modeling sequential data.⁸⁸ It assumes that the system consists of a sequence of unobservable hidden states and associated observable emissions. The core idea is that the hidden states follow the Markov property (the current state depends only on the previous state), with transition probabilities defined by a state transition matrix, while each hidden state generates an observation according to an emission probability matrix. A key feature of HMMs is that the states are not directly observable and must be inferred from the observed sequence. HMMs are widely used in tasks such as speech recognition, natural language processing, and biological sequence analysis, with classic algorithms including the forward-backward algorithm,

Viterbi decoding, and Baum–Welch parameter learning. HMM, as a powerful tool for executing various audio recognition tasks, offers unique advantages in acoustic-based UAV detection. For instance, Guo *et al.*⁸⁹ proposed an acoustic-based system for detecting, classifying, and tracking illegal drones using a circular microphone array and beamforming techniques. Their system employs switched beamforming for initial scanning, an HMM for sound classification, and adaptive recursive least squares (RLS) beamforming for tracking. Simulations demonstrate high accuracy in both ideal and noisy environments, with effective performance even in the presence of interference. In response to the potential threat posed by small uncrewed aerial vehicles (UAVs) in security-sensitive areas, Shi *et al.*⁹⁰ designed a UAV sound recognition method based on MFCC feature extraction and HMM classification, which can capture the temporal characteristics of UAV sounds even in noisy environments. The limitation of HMM lies in its strict Markov assumption (the current state only depends on the previous state), which may not fully capture the complex distance dependencies in drone acoustic signals and may be affected when dealing with complex background noise and multi-source reverberation.

B. Deep learning-based methods

Deep learning has garnered significant attention in the recognition of UAV acoustic signals in recent years. Due to its sophisticated learning mechanisms and robust inference capabilities, it is capable of uncovering more detailed feature information from large datasets of acoustic signals, often yielding superior performance compared to traditional methods, particularly in multi-source acoustic scenarios. The deep learning approaches currently emerging in drone acoustic detection tasks mainly revolve around two core technologies: CNNs and RNNs.

1. Convolutional neural network (CNN)

CNNs are a specialized type of deep neural network known for their ability to extract local features, especially effective in handling data with grid-like structures such as time series or image data. Hence, they are highly suitable for processing acoustic signals and analyzing spectral features. Leveraging CNNs to understand the intricate patterns of UAV acoustic signals facilitates high-precision detection and classification of UAVs.⁹¹ This has significantly advanced the field of acoustic-based UAV detection, where typical CNNs and their variants, such as Residual Network (ResNet)⁹² and Visual Geometry Group (VGG),⁹³ have demonstrated successful implementations.

a. Fundamental principles of CNN. As depicted in Fig. 6, a typical CNN architecture usually comprises multiple processing layers, mainly including convolutional layers, pooling layers, fully connected layers, activation functions, and an output layer. The fundamental structure of a CNN is composed of alternating pairs of convolutional layers and pooling layers.⁹⁴ The convolutional layers employ filters to convolve the entire input space, detecting local feature connections from the previous layer and passing them to the next layer. Meanwhile, the pooling layers calculate the maximum or average values within local regions of the feature map, reducing the data dimensions and simplifying the computation process. Acoustic signals are typically sampled as one-dimensional amplitude sequences over time, allowing for the use of one-dimensional

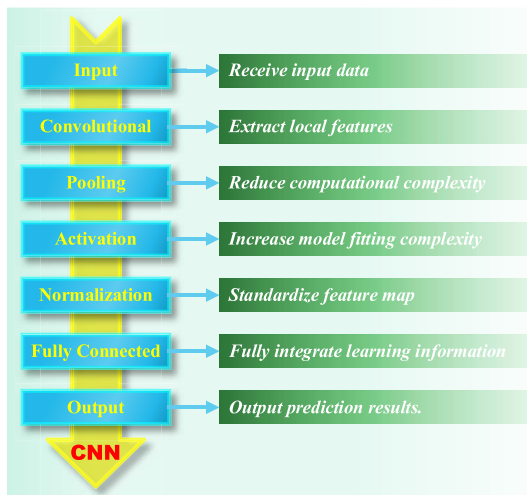


FIG. 6. Typical composition of a CNN architecture.

convolution along the time axis to capture signal patterns. However, to more effectively and comprehensively extract the spectral features of acoustic signals, two-dimensional convolutions are more widely adopted as the core architecture of CNNs. While one-dimensional CNNs generally involve lower computational costs, two-dimensional CNNs are more adept at extracting complex features. The output values of these two convolution operations are as follows:

$$y(t) = \sigma \left[\sum_{k=0}^k x(t+k) \cdot w(k) + b \right], \quad (12)$$

$$y(i,j) = \sigma \left[\sum_{m=0}^m \sum_{n=0}^n x(i+m, j+n) \cdot w(m,n) + b \right], \quad (13)$$

where $y(t)$ represents the output value of the one-dimensional convolution operation at time t , while $y(i,j)$ denotes the output value of the two-dimensional convolution operation at the position (i,j) in the feature map; σ refers to the activation function; b represents the bias term; k indicates the length of the filter; w is the weight of the filter; and m and n correspond to the width and height of the convolution kernel, respectively.

CNNs enable end-to-end processing, extracting features directly from the time–frequency spectrum of raw acoustic signals, and are exceptionally proficient at uncovering local frequency domain features. In addition, by integrating multi-scale features through deep networks, noise resilience is enhanced and can be expanded further through the integration of transfer learning. Currently, this method represents the prevailing deep learning approach for acoustic-based UAV detection, resulting in numerous successful applications. For example, Jiao *et al.*⁹⁵ proposed a lightweight CNN named ADS-CNN for recognizing the flight attitude of quadrotor UAVs based on audio features. By fusing MFCC and STFT features extracted from a self-recorded UAV audio dataset and integrating an attention mechanism and depthwise separable convolutions, the method achieves high recognition accuracy with reduced model complexity. Experimental results show that ADS-CNN outperforms

traditional models in both accuracy and efficiency. Agarwal *et al.*⁹⁶ developed an acoustic-based drone identification system using a CNN model trained on spectrograms of drone audio. The model achieved 95.83% accuracy in distinguishing drone sounds from environmental noise, demonstrating effectiveness in detecting drones via CNN for their acoustic signatures.

These studies have also revealed some drawbacks of CNNs, including high computational demands, extensive data annotation requirements, limited interpretability, and weak generalization capabilities.

b. *Residual network (ResNet)*. As the depth of the CNN increases, it can lead to issues such as gradient decay or even vanishing gradients, making it difficult for the model to effectively update and learn useful data features. As a solution, residual blocks⁹⁷ were incorporated into CNNs, resulting in the ResNet variant, which successfully resolves the problems mentioned above. The residual blocks in ResNet utilize skip connections, allowing the input to bypass the convolutional layers and be directly added to the output, forming a residual. This can be represented as follows:

$$Y(x) = F(x) + x, \quad (14)$$

where x stands for the input data, Y indicates the output result, and F represents the features obtained by applying several convolutional layers to the input x .

ResNet allows gradients to be directly backpropagated through residual connections, thereby increasing network depth and strengthening deep feature extraction capabilities and noise resistance. The convergence speed and resistance to overfitting have also improved. It has attracted attention in some drone acoustic detection studies. For example, Lei *et al.*⁹⁸ proposed a two-stage UAV noise removal method based on a deep ResNet to improve speech intelligibility in UAV noise environments. They used a ResNet with four residual blocks to estimate the spectral gain function and applied it to noisy speech, effectively removing drone noise and improving speech quality. Liu *et al.*⁹⁹ designed an improved lightweight ResNet deep learning model to address the recognition challenges caused by the spectral overlap of drone sounds and background noise in environments with low SNR. This model reduces the number of layers and integrates channel and spatial attention mechanisms, significantly improving feature extraction performance under low SNR conditions while reducing the model's parameters.

However, the ResNet series models discussed in these studies still have limitations, including a large number of parameters, high computational costs, and insufficient sensitivity to dynamic temporal features.

c. *Visual geometry group (VGG)*. VGG is also a variant of CNN, which adopts a modular design by combining convolutional layers, activation functions, and pooling layers into standardized VGG blocks and stacking them repeatedly.¹⁰⁰ This allows for flexible adjustments in network depth and width. VGG increases the network depth while maintaining the receptive field, making it highly suitable for progressively extracting high-frequency harmonics and temporal continuity from time–frequency data, thereby capturing more complex data patterns. VGG exclusively uses small convolutional kernels, making it simple to extend and replicate. It is frequently pre-trained to establish weights, which are then transferred

to a wide range of tasks to serve as a feature extractor. Nonetheless, due to the large size of its network and high computational costs, it necessitates substantial data and computational resources to complete the training process. VGG has significant potential in acoustic feature extraction and transfer learning for UAV detection, particularly suited for fine feature mining and multi-scale analysis in medium- to small-scale datasets, resulting in several representative application outcomes. For instance, Anidjar *et al.*¹⁰¹ proposed a real-time acoustic anomaly detection framework for UAVs, combining a Transformer-based model with a compressed VGG-inspired CNN to analyze drone-emitted sounds. Their method addressed reliability challenges in UAVs by offering a low-cost, microphone-based solution for real-time fault detection in noisy environments. El-Latif *et al.*¹⁰² presented a novel deep learning-based approach that integrates Mel-spectrogram analysis and the VGGish transfer learning model to accurately detect drone payload weights from acoustic signals, achieving an impressive accuracy of 99.98%.

Despite its potential, VGG is constrained by computational cost and network depth, which may cause it to underperform compared to ResNet in real-time or highly detailed classification tasks. In real-world applications, a trade-off between accuracy and efficiency must be considered, along with the characteristics of the data.

2. Recurrent neural network (RNN)

RNNs are effective for processing sequential acoustic data in UAV detection due to their ability to model temporal dependencies. RNN-based systems can operate in real-time, making them suitable for dynamic environments.⁴⁶ By analyzing acoustic patterns over time, RNNs can distinguish drone noise from background interference. Unlike traditional methods, RNNs automatically learn features from raw audio signals, improving detection accuracy. RNN variants, Long Short-Term Memory (LSTM)¹⁰³ and Gated Recurrent Unit (GRU),¹⁰⁴ address the issue of vanishing gradients, enabling robust modeling of long-range dependencies. Applications include surveillance, unauthorized drone identification, and collision avoidance.

a. Fundamental principles of RNN. RNNs are a type of neural network specifically designed for processing sequential data,¹⁰⁵ comprising the input, hidden, and output layers. The key feature of RNNs is that the hidden state h_t at the current time step is determined by both the input x_t and the hidden state h_{t-1} from the previous time step, i.e., $h_t = f(x_t, h_{t-1})$. This recursive connection enables the network to unfold over time and capture temporal dependencies. The hidden state h_t of RNNs is typically a high-dimensional vector that encodes historical information, thereby overcoming the limitations of feedforward neural networks in directly processing variable-length sequences. If the input sequence $X = [x_1, x_2, \dots, x_t]$ at state t is given, the output state y_t at time t can be calculated as

$$h_t = \sigma(W_{xh}x_t + W_{hh}h_{t-1} + b_h), \quad (15)$$

$$y_t = W_{hy}h_t + b_y, \quad (16)$$

where W_{xh} , W_{hh} , and W_{hy} denote the weight matrices; b_h and b_y indicate the bias vectors; σ refers to a nonlinear activation function, generally represented by hyperbolic tangent (tanh), rectified linear

unit (ReLU), logistic function (sigmoid), etc. The state of the hidden layer h_t depends on the previous state h_{t-1} and the input x_t . Essentially, h_t and h_{t-1} act as memories and influence the network's output y_t .

RNNs are effective for acoustic-based UAV detection due to their ability to model temporal dependencies in sound sequences, enabling robust noise filtering and real-time processing. For example, to address the threat of malicious use of commercial amateur drones, Jeon *et al.*¹⁰⁶ proposed a detection method based on RNN and verified its applicability in real-time detection systems using a real-world dataset. Kim *et al.*¹⁰⁷ designed a distributed drone detection system and compared the detection performance of three deep learning models—RNN, CNN, and Dense—on MFCC and Mel spectrogram features. RNN was proven to be the optimal network, achieving an accuracy rate as high as 96%. However, RNNs face challenges such as high computational costs, vanishing gradients in long sequences, and sensitivity to environmental variability in training data.

b. Long short-term memory network (LSTM). LSTM¹⁰⁸ is a special type of recurrent neural network designed to address the issues of gradient vanishing and gradient explosion that arise when traditional RNNs process long sequence data, thereby enabling it to learn and remember long-term dependencies effectively. LSTM achieves this by replacing the standard layers of the neural network with memory units that retain long-term information while filtering out irrelevant short-term information. The specific structure is shown in Fig. 7(a), which primarily includes the cell state (long-term memory unit c_t and short-term memory unit h_t), input gate i_t , output gate o_t , and forget gate f_t . The mathematical representation is as follows:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i), \quad (17)$$

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f), \quad (18)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o), \quad (19)$$

$$h_t = o_t \cdot \text{Tanh}(c_t), \quad (20)$$

$$\widehat{c}_t = \text{Tanh}(W_{xc}x_t + W_{hc}h_{t-1} + b_c), \quad (21)$$

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \widehat{c}_t, \quad (22)$$

where W and b are the weight matrix and bias, x_t is the input sequence data, σ is a logical sigmoid activation function, $\widehat{c}_t$ is an intermediate variable, and Tanh is the activation function.

With its powerful sequence modeling capabilities, LSTM can capture complex temporal patterns and contextual information, thereby improving model performance and enabling it to perform well in a variety of drone detection tasks. For example, Utebayeva *et al.*¹⁰⁹ first attempted to use a stacked bidirectional LSTM network with two hidden layers for drone acoustic detection, aiming to achieve a multi-label classification detection system. The experimental results show that the model can classify various drone sounds with a high accuracy of 94.02%. Li *et al.*¹¹⁰ addressed drone

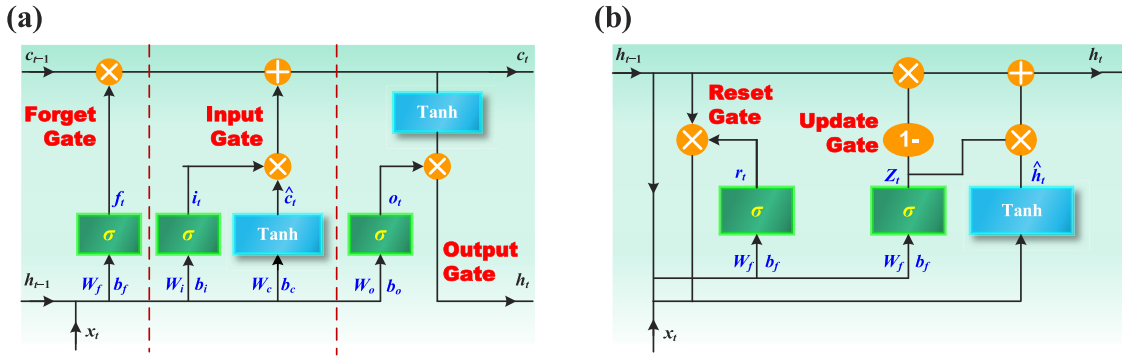


FIG. 7. Network architecture: (a) LSTM and (b) GRU.

intrusion events by combining LSTM, CNN, and self-attention mechanisms in a deep learning framework, effectively distinguishing drone sounds from wind events with 100% accuracy. Even in complex multi-source threat environments, the system maintained a high recognition accuracy rate of 96.25%. However, LSTM also has some inherent challenges. It cannot eliminate gradient issues, especially when dealing with extremely long sequences, and its complex internal structure significantly increases training costs.

c. Gated recurrent unit (GRU). GRU¹¹¹ is a variant of RNN, which, such as LSTM, is designed to solve the long-term dependency problem of RNN. Both use gating mechanisms to control information flow. Their structures are shown in Fig. 7(b). In comparison, GRU has a simpler structure and fewer parameters, which means that it requires fewer computing resources during training and trains faster. Its core lies in two key gating units: the update gate z_t and the reset gate r_t . These gating units control the flow of information through a learning process, as detailed as follows:

$$z_t = \sigma(W_{xz}x_t + W_{hz}h_{t-1}), \quad (23)$$

$$r_t = \sigma(W_{xr}x_t + W_{hr}h_{t-1}), \quad (24)$$

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \hat{h}_t, \quad (25)$$

$$\hat{h}_t = \text{Tanh}(W_{xh}x_t + W_{rh}(r_t \cdot h_{t-1})), \quad (26)$$

where h_t represents the unit state, W represents the weight matrix, σ and Tanh are the activation functions, and $\hat{h}_t$ is the intermediate variable.

The GRU reset gate can help the model weaken unimportant or potentially noisy information, enabling it to focus more on key features in the signal. This is particularly important for drone acoustic detection in complex acoustic environments, making it an efficient and popular choice. For example, Utebayeva *et al.*²⁷ conducted an in-depth study of various drone sound recognition systems based on recurrent neural network architectures, where GRU achieved the best performance. Notably, they also focused explicitly on drone

payload conditions, achieving an accuracy rate of up to 98% for payload status classification across multiple categories, including loading, unloading, and background noise. Another study⁴³ successfully validated the feasibility of determining the discrete distance range of UAVs using sound signals. Leveraging GRU's robust temporal modeling capabilities, the system achieved classification accuracy rates of 94% and 98% for UAV sounds classified at intervals of 10 and 15 m, respectively, within a range of 5–50 m. However, GRU is more sensitive to parameter tuning, such as learning rate, due to its compact structure, and it still faces challenges, such as limited context scope constrained by the depth of the recurrent layers and relatively poor interpretability.

3. Other deep learning methods

In addition to pure CNNs and RNNs, various innovative deep learning methods have emerged in the field of drone acoustic detection. These methods enhance detection performance by integrating the advantages of different architectures or introducing new mechanisms, greatly enriching the technical means of drone acoustic detection and promoting the development of this field toward greater efficiency and robustness.

a. Convolutional recurrent neural network (CRNN). CRNN¹¹² is a hybrid deep learning framework that cleverly combines the advantages of CNN in spatial feature extraction and the capabilities of RNN in sequence modeling. This collaborative approach enables CRNN to demonstrate unique performance in processing time-series audio data with local features, achieving outstanding results in drone acoustic detection tasks. For example, Kashyap *et al.*¹¹³ proposed an innovative deep learning architecture based on CRNN, achieving a detection accuracy of up to 99.30% for small drones within a 70-m airspace control range, highlighting the significant potential of CRNN in this field. Yang *et al.*¹¹⁴ utilized a CRNN network to address the dynamic evolution of drone acoustic features (fundamental frequency and harmonic structure). This approach inherits the strong ability of CNNs to extract local features in image-like spectra while effectively capturing the dynamic patterns of drone audio using RNNs, thereby achieving a comprehensive understanding of the spatiotemporal dynamics of drone acoustic signals. However, the high demand for data volume and

limitations in handling ultra-long sequences and interpretability are challenges that CRNN must address in practical applications.

b. Concurrent neural network (CoNN). CoNN¹¹⁵ is a parallel neural network architecture that has demonstrated unique potential in drone acoustic detection. Its core idea is to enhance the network's feature learning capabilities and information integration efficiency through parallelization and multipath processing. This means that multiple independent computing units within the network simultaneously process different parts or aspects of the input data. This design philosophy endows it with robust flexibility and scalability, enabling it to handle complex, multi-dimensional audio data rich in contextual information with distinct advantages. It has been successfully applied in drone acoustic detection tasks. For example, Dumitrescu *et al.*⁵¹ developed a drone acoustic detection system based on the CoNN architecture, combining three different types of competitive networks: a multilayer perceptron (MLP), a time-delay neural network (TDNN), and a self-organizing map (SOM) mapping. Independent training is used to solve different subproblems. From the experimental results, CoNN exhibits better generalization ability and faster convergence, resulting in higher accuracy. The theoretical and experimental results prove their feasibility and provide new ideas for solving complex audio signals. It is worth noting that the parallel capability of CoNN makes it very suitable for detecting and tracking multiple drone targets. However, the high computational overhead and design complexity remain challenges and are also the key to its development. In particular, CoNN subnetwork parallelization will significantly increase inference memory on the one hand and complicate architecture threshold and weight tuning on the other.

c. Transformers. In recent years, deep learning technology has made significant progress in various fields, and the Transformer model is undoubtedly one of the most notable innovations in the field of artificial intelligence. Vaswani *et al.*¹¹⁶ first proposed the Transformer architecture, which completely abandons recurrent and convolutional structures. Leveraging its unique self-attention mechanism and parallel computing capabilities, it demonstrates exceptional performance in processing long sequence data, successfully replacing recurrent networks to a certain extent and introducing a new paradigm for audio processing. A typical Transformer architecture primarily consists of an encoder-decoder structure, a self-attention mechanism, multi-head attention, position encoding, and a feedforward neural network. The self-attention mechanism is its core component, described as a mapping of a set of query vectors and key vectors to an output vector. First, the dot product between the query and key vectors is calculated and scaled. Then, the similarity matrix is normalized using *softmax* normalization to ensure compatibility between them. The output vector is the weighted sum of the value vectors, with the scoring formula expressed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right)V, \quad (27)$$

where $Q \in \mathbb{R}^{n \times d_Q}$ denotes the query matrix, $K \in \mathbb{R}^{n \times d_K}$ indicates the key matrix, $V \in \mathbb{R}^{n \times d_V}$ represents the value matrix, d_Q , d_K , and d_V are their dimensions, n refers to the sequence length, and $\sqrt{d_K}$ denotes the scaling factor.

Through the self-attention mechanism, Transformers can capture complex time-frequency dependencies in audio signals, which is advantageous for drone detection in time-varying and complex acoustic environments. For example, Anidjar *et al.*¹⁰¹ introduced the Transformer architecture into drone acoustic anomaly detection to handle audio recognition in more challenging environments. By leveraging Self-Supervised Learning (SSL) pre-training, they significantly reduced the requirement for drone acoustic data, demonstrating the feasibility of Transformers for small-sample, edge-side drone acoustic detection. Berg *et al.*¹¹⁷ compared the classification performance of a fine-tuned Audio Spectrogram Transformer (AST) architecture and a custom CNN in a small-sample scenario of 4500 s and nine drone classes. The results showed that although AST had slightly lower overall accuracy than CNN, its accuracy increased with the scale of the data, demonstrating scalability with the potential to outperform CNN as the data grow. Transformers have demonstrated their effectiveness in multiple detection tasks. However, they still face challenges such as high computational resources and data requirements, high computational overhead for long sequences, and a lack of local inductive bias.

C. Comparison of various learning methods in acoustic-based UAV detection

1. General detection performance evaluation metrics

To accurately evaluate the performance of different detection systems and algorithms, a series of scientific and objective evaluation metrics¹¹⁸ is required, aiming to provide a comprehensive reference for the design, optimization, and evaluation of UAV acoustic detection systems. The performance evaluation metrics for drone acoustic detection classification primarily include accuracy (*Acc*), precision (*P*), recall (*R*), and F1 score (*F1*). Accuracy measures the correctness of the model's performance across all samples, indicating the model's overall ability to classify samples correctly. Precision focuses on the model's ability to avoid false positives, i.e., the proportion of audio signals correctly identified as drones. Recall is the ratio of correctly predicted drone samples to all drone samples. The F1 score comprehensively considers precision and recall, providing a balanced performance metric. The formula is expressed as

$$\begin{cases} \text{Acc} = \frac{TP + TN}{TP + TN + FP + FN}, \\ P = \frac{TP}{TP + FP}, \\ R = \frac{TP}{TP + FN}, \\ F1 = 2 \times \frac{P \times R}{P + R}, \end{cases} \quad (28)$$

where *TP* represents the true positives, indicating the number of correctly classified results as drones; *FP* denotes the false positives, indicating the number of incorrectly classified results as drones; *TN* indicates the true negatives, indicating the number of correctly classified results as non-drones, and *FN* refers to the false negatives, indicating the number of incorrectly classified results as non-drones.

In addition to the above classification performance indicators, drone acoustic detection may also involve important indicators

such as detection distance, signal-to-noise ratio (SNR), and processing time to evaluate the practicality and efficiency of the system comprehensively.

2. Comparison of traditional machine learning methods

Traditional machine learning methods are mostly simple in principle and played an important role in early drone acoustic detection research, demonstrating different characteristics. SVM is a typical binary classification model that introduces kernel functions to map the original feature space to a high-dimensional space, enabling the handling of nonlinear classification problems in complex drone audio. However, SVM is highly sensitive to parameter settings and kernel function selection, requiring careful tuning. KNN is a non-parametric supervised learning algorithm with an intuitive algorithm concept and natural support for multi-classification problems, without the need for additional combination strategies such as SVM. However, it is a “lazy learning” algorithm that requires traversing all training samples for each prediction, resulting in high computational overhead during the prediction phase. MLP is the most basic feedforward neural network, which can capture complex patterns in drone sounds through multiple layers of nonlinear transformations and has a structure that is easy to integrate into existing deep learning frameworks. However, in practical applications, it still faces challenges such as overfitting, long training times, and poor interpretability. HMM is a statistical model particularly suited for processing time-series data such as drone audio, identifying the most likely sequence of hidden states that generated the sequence to capture dynamic characteristics and temporal dependencies. This is crucial for detecting time-varying drone sounds. However, its heavy reliance on independence assumptions limits its discriminative ability during detection.

In conclusion, traditional machine learning methods are generally simple in principle and have strong interpretability. Although they perform less effectively than deep learning models in processing complex and dynamic drone acoustic signals, they are not entirely without value. In particular, in scenarios with resource constraints, limited data, or high requirements for model interpretability, they can still leverage their unique advantages and collaborate with deep learning methods to enhance the performance and practicality of detection systems.

3. Comparison of deep learning methods

In recent years, with the rapid development of deep learning, it has become the core driving force in the field of drone acoustic detection. Among these, CNN-based structures excel at capturing local spatiotemporal features, making them suitable for drone sound recognition based on acoustic spectrum maps, particularly in feature extraction and classification tasks. Its variant, ResNet, effectively mitigates the gradient vanishing problem in deep networks by introducing residual connections. VGG, with its stacked structure, enhances the network's nonlinear expression capabilities while maintaining feature extraction depth. However, these architectures have certain limitations in capturing time-dependent features, computational resource requirements, and reliance on large-scale datasets. Given the inherent temporal characteristics of acoustic signals, RNNs and their variants (such as LSTM and GRU) have played a crucial role in drone acoustic detection, particularly in processing

audio temporal information and long-range dependencies, making them highly suitable for scenarios requiring the capture of dynamic changes in drone sounds. However, the inherent sequence processing mechanisms of such models limit their parallel computing capabilities, resulting in relatively low training efficiency, especially when handling ultra-long sequences. To fuse the advantages of different networks and overcome their limitations, hybrid neural networks have emerged. For example, CRNN combines the advantages of CNN and RNN to fully capture the spatial and temporal dependency information of drone sounds, demonstrating excellent performance in various acoustic detection tasks. CoNN aims to improve model efficiency and performance through parallel processing and information fusion. Although its research in the field of drone acoustic detection is relatively limited, its potential for multi-scale feature processing and multi-sensor fusion deserves further exploration.

In addition, the introduction of the emerging Transformer architecture has brought breakthroughs to drone acoustic detection. Its parallel computing capabilities and exceptional ability to capture long-range dependencies make it highly promising for processing large-scale acoustic data and complex scenarios. However, the large number of parameters and computational requirements limit its direct deployment in resource-constrained environments.

Overall, various deep learning models have their strengths in drone acoustic detection, and the choice of an appropriate model depends on the specific application scenario, data characteristics, and requirements for computational resources and real-time performance. In the future, with the development of drone edge computing and embedded devices, developing lightweight, high-performance deep learning models will become an important trend. Given the lack of drone sound datasets, researching meta-learning, contrastive learning, and generative adversarial networks to utilize limited datasets efficiently is expected to play a key role in this field. In addition, efforts to enhance model transparency and interpretability and to understand the decision-making processes of deep networks will help optimize models and improve their robustness in complex real-world environments.

4. Benchmark-based comparison of detection performance for various learning methods

To evaluate various methods in drone acoustics, we conducted drone acoustic detection experiments on a public dataset,⁴⁰ which contains two categories of audio samples: drone and non-drone, with a total duration of 1332 s for each category. All audio files are in WAV format, with a sampling rate of 16 kHz and a bitrate of 16 kbps. UAV audio was obtained by recording flight and hovering sounds and overlaying real background noise. In contrast, non-UAV audio includes various noise types such as urban noise, helicopters, airplanes, birds, thunderstorms, storms, and rain, with a signal-to-noise ratio ranging from -32 to -3 dB. Each sample is 1 s long, and the dataset consists of 2664 samples, which are divided into a training set (2131 samples) and a test set (533 samples) in an 8:2 ratio. The performance metrics considered include accuracy (Acc), F1 score (F1), precision (P), recall (R), and test time. The experimental setup utilized an Nvidia GeForce RTX 4090 Ti graphics card, a 3 GHz Intel CPU, and 256 GB of memory. The network parameters were configured as follows: epoch = 30, batch size = 16, and learning rate = 0.000 01.

TABLE III. Results of different networks in acoustic-based UAV detection tasks.

Methods	Networks	Metrics				
		Acc (%)	F1 (%)	P (%)	R (%)	Test time (s)
ML	SVM	89.32	88.85	89.32	88.38	0.73
	KNN	87.54	86.46	87.32	85.62	0.61
	MLP	91.87	90.20	89.46	90.95	0.82
	HMM	78.96	78.50	78.96	78.04	1.02
DL	CNN	94.56	93.84	93.12	94.58	3.49
	ResNet	95.98	95.71	95.74	95.69	4.13
	VGGish	95.34	95.38	95.34	95.36	4.87
	RNN	88.76	88.27	88.76	87.78	2.92
	LSTM	95.23	94.65	94.32	94.98	3.61
	GRU	94.89	93.73	94.56	92.91	3.43
	CRNN	96.12	95.34	95.56	95.13	5.17
	CNN-LSTM	96.87	96.88	96.87	96.89	5.73
	Transformer	96.43	96.43	95.43	96.44	6.84

Table III presents the detection results for various networks. Among the ML methods, MLP achieved the best performance using nonlinear mapping (Acc 97.87% and F1 90.20%), followed by SVM and KNN. HMM performed significantly worse than other methods (Acc 78.96 and F1 78.50), which may be due to its limitation under the Gaussian assumption, making it less capable of capturing the complex harmonic and time-varying characteristics of drone acoustic signals. It is worth noting that these algorithms were tested within less than 1 s, with extremely low computational overhead, making them suitable for resource-constrained early warning or rapid prototyping scenarios. Among DL models, CNN achieved a significant leap in performance with its robust feature learning capabilities, improving accuracy to 94.56%. Its variants, ResNet and VGGish, further improved feature learning efficiency and effectiveness through deeper network structures and specific designs. Compared to other DL methods, the baseline RNN performs poorly with an accuracy rate of only 88.76%. This may be due to its unstable training and susceptibility to gradient vanishing, resulting in insufficient expressive power and noise robustness. LSTM and GRU effectively address the long-range dependency issue of RNN through gating mechanisms, achieving significant performance improvements (LSTM Acc 95.23% and GRU Acc 94.89%). CRNN and CNN-LSTM further integrate convolutional and recurrent structures, achieving the best performance, with CNN-LSTM reaching the highest performance in this experiment (Acc 96.87% and F1 96.88%). Transformer models leverage their self-attention mechanism to focus on key information in drone sounds, achieving performance on par with fusion-based models and demonstrating strong potential for handling global information and long-range dependencies. However, their testing time is the longest among all models, reaching 6.84 s.

In summary, DL models demonstrate outstanding performance in uncrewed vehicle acoustic detection tasks, significantly outperforming traditional ML methods. However, while improving detection performance, they also increase computational complexity, resulting in significantly longer testing times compared to ML-based

methods. To achieve broader and more reliable practical applications, future research should go beyond the pursuit of high accuracy and instead focus on model lightweighting and efficient deployment, robustness enhancement in complex environments, model interpretability, multimodal fusion, and collaborative perception.

V. ACOUSTIC-BASED UAV LOCALIZATION

The fundamental principle of UAV acoustic localization involves estimating the direction-of-arrival (DOA)¹¹⁹ through advanced signal processing techniques that analyze sound wave propagation characteristics across sensor arrays, subsequently determining the spatial coordinates via geometric triangulation. Acoustic source localization techniques can be systematically categorized into three principal methodologies based on their computational foundations: time-difference-of-arrival (TDOA) estimation,¹²⁰ beamforming-based approaches,¹²¹ and high-resolution spectral analysis techniques.¹²² Table IV summarizes the existing studies on UAV acoustic source localization.

A. Array signal model

The propagation of UAV acoustic signals through the medium can be modeled as either a near-field or far-field scenario,¹³⁰ depending on the distance between the acoustic sensor array and the UAV. The acoustic field of a UAV is modeled as near-field when it satisfies the following criterion; otherwise, it is treated as a far-field model:

$$L < \frac{2D^2}{\lambda_{\min}}, \quad (29)$$

where D stands for the sensor array's aperture, $\lambda_{\min}$ denotes the shortest wavelength of the signal, and L specifies the range from the UAV sound source to the array.

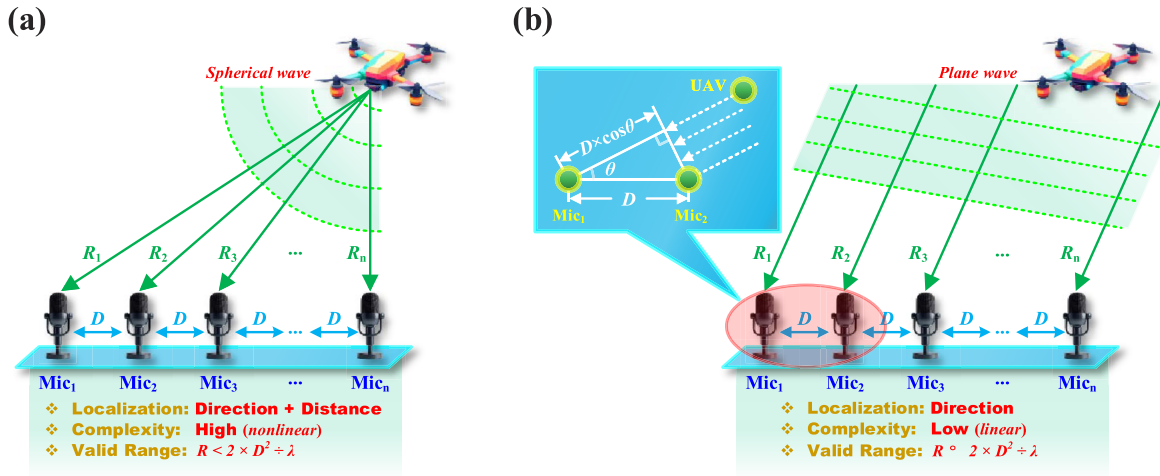
As illustrated in Figs. 8(a) and 8(b), for short-range sound sources, the near-field model assumes spherical wavefronts impinging on the sensor array. It considers both the amplitude and phase differences of the sound waves, thereby offering a more accurate approximation of real acoustic behavior. Conversely, the far-field model assumes that the sound source is far from the array, leading to minimal differences in the angles of incidence. As a result, signal phase information is typically ignored, and wave propagation is modeled using plane waves. The variation in incident angles at the array elements between near-field and far-field conditions makes selecting a suitable model crucial for accurate localization. Typically, UAV monitoring environments meet the criteria for the far-field model.

B. Time-difference-of-arrival (TDOA) localization technology

TDOA technology is a time-domain approach for estimating the DOA of signals, commonly applied in acoustic source localization.¹³¹ Utilized initially for the localization of radio and acoustic signals, TDOA technology has gained widespread application in acoustic-based UAV detection with the development of UAV detection and counter-UAV technologies, especially in precise UAV localization using microphone arrays. This technique works by measuring the time delay of the same acoustic signal reaching various array elements and applying triangulation methods and sound

TABLE IV. Summary of related work on acoustic localization of UAVs.

References	Method	Microphone array	Sensor	Function	Range
82	TDOA	Tetrahedron	CHZ-213	Localization	100 m
34	TDOA	Tetrahedron	FOAS	Localization and tracking	Long-distance
31	MUSIC	Spherical	MP-40	Localization (low-SNR)	20 m
123	TDOA	Triangle	MEMS	Localization	Area of $15 \times 15 \text{ m}^2$
124	MUSIC	Spherical	MEMS	Tracking (multi-UAVs)	60 m
125	ESPRIT	Spherical and circular	MEMS	Localization and interception	None
126	Beamforming	Two-dimensional plane	MEMS	Localization and tracking	11 m
127	TDOA	Tetrahedron	130F22	Localization	Area of $10 \times 10 \text{ m}^2$
128	TDOA	Tetrahedron	Silicon microphone	Localization	19.18 m
129	Beamforming	Tetrahedron	MPA-416	Localization and tracking	10 m

**FIG. 8.** Acoustic-based UAV localization model: (a) Near-field and (b) Far-field.

propagation models to infer the three-dimensional location of the source. Figure 8(b), for instance, depicts a schematic representation of a one-dimensional linear array used to estimate a single drone sound source in the far-field condition. In this case, the relationship between the time delay of two microphones and the array's geometry is established as follows:

$$\tau = \frac{D \times \cos \theta}{c}, \quad (30)$$

where τ indicates the time delay, D stands for the inter-element distance, c represents the sound speed in air, and θ is the direction of arrival angle of the sound wave to be estimated.

To estimate the time difference of arrival τ between the signals $x_1(t)$ and $x_2(t)$ from two microphones, the cross correlation function is commonly used as a criterion, expressed as

$$R_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t)x_2(t-\tau)dt, \quad (31)$$

$$\hat{\tau} = \arg \max R_{12}, \quad (32)$$

where $R_{12}(\tau)$ denotes the cross correlation function, and $\hat{\tau}$ represents the estimated delay. When we shift the later-arriving signal $x_2(t)$ along the time axis, the point where it most closely matches the earlier-arriving signal $x_1(t)$ overall (the position where the cross correlation reaches its maximum value) corresponds to the maximum value. By locating this maximum value, we obtain the estimated delay. To obtain more detailed spatial information about the UAV, a more intricate array structure is necessary to create a broader spatial geometric relationship, thereby enabling the accurate determination of the UAV's location. Given that time delay estimation errors directly influence the accuracy of UAV localization, a key research challenge is developing methods to rapidly and accurately determine the time delays between signals. As signal processing technologies have advanced, time delay estimation methods now include prominent approaches such as the generalized cross correlation (GCC) algorithm,¹³² adaptive least-squares filter algorithm,¹³³ and quadratic correlation algorithm.¹³⁴ Among these methods, the GCC algorithm has emerged as one of the most commonly used methods, being widely applied in UAV acoustic detection.

TDOA acoustic source localization relies on the precise synchronization of multiple sensors to locate UAVs, and a variety of

significant applications have emerged in recent years. For instance, Shi *et al.*⁸² developed a TDOA algorithm based on a tetrahedral microphone array, utilizing a Bayesian filter and a Gaussian mixture model to effectively address multipath interference, achieving a detection range greater than 100 m in real-world settings. Fang *et al.*³⁴ investigated a UAV detection system based on optical fiber acoustic sensors, integrating distributed acoustic sensing (DAS) technology with the TDOA localization method. By exploiting the time differences across various optical fiber sensing points, the system enables real-time 3D positioning and precise 3D tracking of the UAV. Accurate time delay estimation algorithms are key to achieving high-precision positioning. Liu *et al.*¹²⁸ focus on improving time delay estimation accuracy in low signal-to-noise ratio (SNR) environments. They utilize a novel weighting function that combines the characteristics of the smoothing coherence transform (SCOT) and phase-hopping transform (PHAT), enabling the algorithm to achieve more accurate time delay estimates under low SNR conditions, with drone sound source localization errors not exceeding 22.1 cm. Lim *et al.*¹²⁷ propose a drone sound source localization method using a distributed dual microphone array and combine it with GCC-PHAT technology to improve peak localization by 2.13 m in environments with a signal-to-noise ratio higher than 0 dB.

At present, TDOA technology reveals great application potential in UAV acoustic detection, and its high accuracy and real-time efficiency strengthen its status as the primary technology for UAV sound source localization. For long-range detection, integrating noise filters or adaptive signal processing technologies holds the potential to further improve localization accuracy in challenging environments. With advancements in sensor networks and edge computing, TDOA plays a key role in enabling efficient real-time

acoustic monitoring of large UAV swarms, while also optimizing computational efficiency and minimizing power consumption.

C. Super-resolution localization technology

1. Multiple signal classification (MUSIC) algorithm

The multiple signal classification (MUSIC)¹³⁵ algorithm, as depicted in Fig. 9(a), is a super-resolution technique that uses eigenvalue decomposition, typically employed to estimate the DOA or time delay differences in signals. In UAV sound source localization, it can pinpoint the direction of sound sources by estimating the DOA from multiple sources. The fundamental principle of MUSIC is to decompose the received signal's covariance matrix into signal and noise subspaces, utilizing the orthogonality of these subspaces to infer the DOA of the signal. Thus, it can be regarded as a subspace algorithm specifically for narrowband signals.

Because of the unique acoustic signals of UAVs, which span a wide frequency range with clear linear components, it is easier to split the signal into different frequency sub-bands in the frequency domain, making it possible to apply narrowband DOA processing. Assuming the measurement array consists of M sensors with a sensor spacing of d , and L is the number of signal sources being observed, the signal received by the array at time t can be considered as a sum of a series of plane waves, which can be represented as follows:

$$X(t) = AS(t) + N(t), \quad (33)$$

where $X(t) = [x_1(t), x_2(t), \dots, x_M(t)]^T$ is the signal vector received by the $M \times 1$ sensors in the array, $S(t)$ represents the vector of source signals with a size of $L \times 1$, $N(t)$ denotes the noise vector of

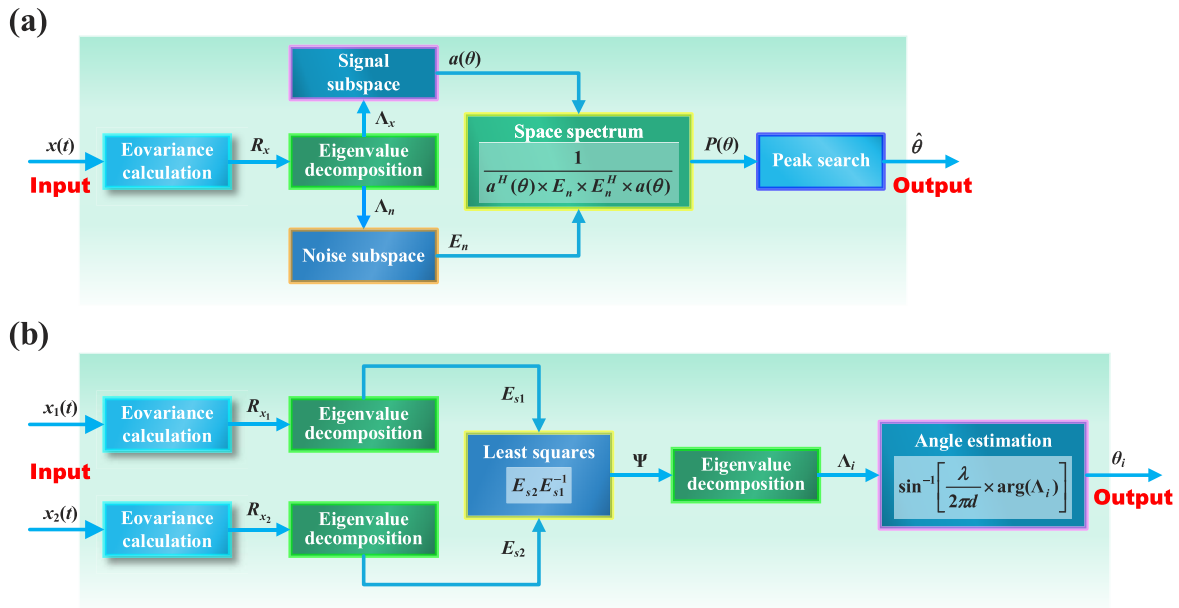


FIG. 9. Algorithm frameworks: (a) MUSIC and (b) ESPRIT.

size $M \times 1$, $A = [a(\theta_1), a(\theta_2), \dots, a(\theta_L)]$ indicates the array steering matrix of dimension $M \times L$, θ is the angle of DOA, and $a(\theta_L)$ is the array steering vector employed to describe the direction of the signal, which can be represented as follows:

$$a(\theta_L) = \left[1, e^{-j \times 2\pi \times d \times \sin \frac{\theta_L}{\lambda}}, \dots, e^{-j \times 2\pi \times (M-1) \times d \times \sin \frac{\theta_L}{\lambda}} \right]^T, \quad (34)$$

where λ refers to the wavelength of the acoustic signal.

Next, the covariance matrix R_x is calculated from the observed signal,

$$R_x = E[X(t) \times X^H(t)] = A(\theta) \times R_s \times A^H(\theta) + \sigma^2 \times I, \quad (35)$$

where $E[\cdot]$ stands for the expectation value, R_s refers to the covariance matrix of the source signals, σ denotes the noise power, and I is the identity matrix.

When the number of sound sources L is fewer than the number of array elements M , we assume that $A(\theta) \times R_s \times A^H(\theta)$ is a full-rank matrix, so the covariance matrix decomposition can be written as follows:

$$R_x = E_s \times \Lambda_s \times E_s^H + E_n \times \Lambda_n \times E_n^H, \quad (36)$$

where E_s is the eigenvector matrix for the signal subspace, linked to the larger eigenvalues Λ_s , and E_n is the eigenvector matrix for the noise subspace, related to the smaller eigenvalues Λ_n .

The key idea behind the MUSIC algorithm is the orthogonality between the signal and noise subspaces. The steering vector $a(\theta_L)$ resides in the signal subspace, meaning it is always orthogonal to the noise subspace. Therefore, the MUSIC algorithm estimates the DOA by constructing a spectral function $P_{\text{MUSIC}}(\theta)$, which is represented as follows:

$$P_{\text{MUSIC}}(\theta) = \frac{1}{a^H(\theta) \times E_n \times E_n^H \times a(\theta)}. \quad (37)$$

The spectral function shows peaks in the signal's direction. By finding these peaks, the DOA angle of the signal can be determined.

Recently, several improvements and extensions have been proposed to enhance the performance of the MUSIC algorithm, such as simplifying the calculation of the spectral function to lower complexity and developing algorithm variants to handle wideband signals. Satish and Medda¹²⁴ designed a spherical harmonic function-based MUSIC algorithm, decomposing the received acoustic signals into spherical harmonic functions and then processing the mutual spectrum matrix of the signals using the MUSIC algorithm, successfully achieving simultaneous tracking of the azimuth angles of two drones. This improvement enhances the applicability of the MUSIC algorithm for processing broadband acoustic data from drones. Wu *et al.*³¹ proposed a MUSIC algorithm based on pseudo-spectrum normalized mean processing for low signal-to-noise ratio environments, eliminating the influence of energy differences among sub-bands on direction of arrival (DOA) estimation. This approach achieves more robust and more precise spatial spectral peaks under low signal-to-noise ratio conditions while effectively suppressing pseudo-peak interference.

To sum up, the MUSIC algorithm can break through the usual resolution limits, making it possible to tell apart multiple signal sources that are very close to each other. More noteworthy is that the

MUSIC algorithm is capable of concurrently estimating the angles of arrival for multiple signals, rendering it suitable for the acoustic source localization of multiple UAVs. Furthermore, it exhibits considerable robustness against Gaussian noise. However, due to the involvement of eigenvalue decomposition of the covariance matrix, the MUSIC algorithm entails a substantial computational load, rendering real-time processing challenging. With algorithmic optimizations and hardware acceleration, MUSIC is expected to continue as a key choice for high-precision UAV localization tasks in the future.

2. Estimation of signal parameters via rotational invariance technique (ESPRIT)

ESPRIT is an efficient algorithm for estimating the DOA of signals.¹³⁶ Its basic principle is based on the translation-invariant property of subarrays within the sensor array, which induces rotational invariance in the signal subspace. Through matrix decomposition and signal subspace methods, it extracts the spatial information of the signal from the received array signals. Compared to the MUSIC algorithm, ESPRIT has the advantage of not requiring spatial search through power spectra, resulting in relatively lower computational complexity and higher efficiency.

As shown in Fig. 9(b), ESPRIT requires the array to be divided into two subarrays. Assuming the sensor positions have a fixed distance d (for instance, the distance between microphones), the geometric structure between the two subarrays remains unchanged, thereby satisfying rotational invariance. The signal model received by the array is defined in the same manner as that of the MUSIC algorithm. After computing the covariance matrix and performing eigenvalue decomposition, the signal subspaces E_{s1} and E_{s2} for the two subarrays are obtained. According to the rotational invariance relationship, there exists a rotation matrix Ψ such that $E_{s2} = E_{s1} \times \Psi$. The diagonal elements of Ψ correspond to the phase delays between the array elements, which can be represented as follows:

$$\Psi = \text{diag} \left[1, e^{-j \times 2\pi \times d \times \sin \frac{\theta_1}{\lambda}}, \dots, e^{-j \times 2\pi \times (M-1) \times d \times \sin \frac{\theta_1}{\lambda}} \right], \quad (38)$$

where λ is the wavelength of the acoustic signal, and θ_i denotes the angle of arrival. The solution for Ψ can be obtained using numerical analysis methods such as least squares or total least squares, and its eigenvalues contain information regarding the signal's angle of arrival.

The phase angle is related to the angle of arrival of the signal as follows:

$$\theta_i = \sin^{-1} \left[\frac{\lambda}{2\pi \times d} \times \arg(\Lambda_i) \right], \quad (39)$$

where $\arg(\Lambda_i)$ indicates the phase angle, and Λ_i represents the eigenvalues of the matrix Ψ .

The ESPRIT algorithm has high application value in the field of drone acoustic positioning due to its advantages of fast calculation speed and high resolution. Hu *et al.*¹²⁵ proposed an improved ESPRIT algorithm based on Toeplitz matrix reconstruction, which fully utilizes array reception data to provide a full-rank equivalent source covariance matrix. In an environment with a signal-to-noise ratio (SNR) of -7.5 dB, the resolution probability exceeds 95%.

Notably, the authors successfully applied this algorithm to a self-developed UAV autonomous interception defense system, demonstrating real-time, precise tracking and interception capabilities for UAV targets in various practical scenarios.

Overall, in DOA estimation, if the application scenario with sufficient computational resources demands a higher resolution, the MUSIC algorithm is a powerful choice. For scenarios requiring real-time processing of large-scale arrays or involving high complexity in angle search, the ESPRIT algorithm is more efficient and suitable due to its spectrum search-free nature. However, it is worth noting that ESPRIT has a strong dependence on the array's geometric structure. If the structure is poorly designed, it may lead to a decrease in noise resistance.

D. Beamforming localization technology

Beamforming is a standard array signal processing technique.¹³⁷ The fundamental concept is to weight and sum the signals from each array element, orienting the array's beam toward the spatial direction with the highest signal energy at a specific time. According to this principle, the steering vector that maximizes the desired signal's output power can be determined, indicating the signal's DOA. By adjusting the weights and phases, the beamforming algorithm is able to dynamically tune its sensitivity and track sound sources from various directions.

In UAV sound source localization, beamforming algorithms are used to determine the direction of each UAV's sound source based on the signals received by the sensor array. By integrating triangulation methods, the precise positioning of multiple UAVs can be achieved, making it highly applicable for complex airspace control and UAV monitoring missions. Existing research results showcase the benefits of beamforming technology in UAV acoustic localization. As an example, Salvati *et al.*¹³⁸ developed a spectral distance response algorithm based on beamforming, enabling efficient localization and enhancement of source signals, effectively identifying different types of sound sources. By incorporating the beamforming algorithm into an anti-UAV system, Cheng *et al.*¹²⁶ enabled UAV sound source localization, overcoming the challenges of visual tracking in adverse environments. Itare *et al.*¹²⁹ proposed a method for locating drones by tracking their acoustic harmonic characteristics in the time–frequency domain. Its core advantage lies in the use of beamforming technology to automatically ignore most of the irrelevant background noise when calculating the direction of arrival (DOA), thereby independently locating the drone on the energy map and significantly improving the signal-to-noise ratio and robustness of the location.

Beamforming sound source localization offers clear benefits in acoustic-based UAV detection, such as boosting signal strength, resisting interference, and enabling the tracking of multiple UAVs, making it ideal for surveillance and safety operations in complex environments. As such, beamforming localization is of significant importance for UAV flight monitoring and intrusion detection systems. Nevertheless, it encounters issues such as multipath interference, high computational demands, and equipment cost. As algorithms continue to improve and hardware performance advances, it is expected to become even more crucial in UAV security tasks.

TABLE V. Advancements in multimodal UAV detection studies.

References	Areas	Task
139	Acoustic, vision, and RF	Detection and localization
91	Acoustic, vision, and radar	Detection
140	Acoustic, vision, and radar	Detection and localization
141	Acoustic and vision	Detection
142	Acoustic and vision	Detection
143	Acoustic, vision, and radar	Detection and tracking
144	Acoustic and RF	Detection
145	Acoustic and vision	Detection and tracking
114	Acoustic and vision	Detection and localization
146	Acoustic, vision, and RF	Detection and tracking

VI. FUTURE AND CHALLENGES

Acoustic-based UAV detection remains in the nascent phase, with numerous technologies still underdeveloped. Furthermore, several open challenges have arisen during its evolution that hinder its operational efficiency, such as susceptibility to strong noise interference, insufficient large-scale datasets, difficulties in detecting and localizing multiple UAVs, and the limited applicability of existing detection systems. To tackle the existing open issues, we have conceptualized a few technological development strategies, which are presented as follows.

A. Multimodal detection models

Given the constraints of individual detection technologies in handling complex real-world environments, acoustic data from various sensor types is incorporated, integrating multiple techniques (e.g., acoustic detection, radar detection, visual detection, and RF detection) into a unified model for enhanced detection capability. As presented in Table V, there are ongoing studies dedicated to this area, with the goal of delivering more comprehensive and efficient solutions for drone security. Future studies in this field will place greater emphasis on data synchronization and integration, coordination between various sensors, improving the capacity to handle large datasets, and the development of more intelligent fusion algorithms. The development of large-scale multimodal models is expected to greatly enhance the advancement of UAV detection.

B. Large-scale proprietary datasets

The primary challenge in developing acoustic-based UAV detection technology lies in the lack of public acoustic datasets, which significantly limits models' training and testing performance. The primary challenge in developing acoustic-based UAV detection technology lies in the lack of public acoustic datasets, which significantly limits models' training and testing performance. In recent years, several representative public datasets have emerged, summarized in Table II. All datasets were collected independently

by researchers, ensuring data authenticity and traceability—a foundation for advancing the field. However, most focus on common consumer-grade and small multi-rotor drones, failing to adequately account for drones in complex real-world environments. This results in insufficient practicality and representativeness of the data, leaving gaps compared to operational requirements. Examples include inadequate coverage of environmental diversity and the absence of unified industry standards for recording equipment, environmental control, and annotation specifications. These issues make it difficult to conduct fair performance comparisons and generalization evaluations between models trained on different datasets. Existing datasets typically capture acoustic signals within tens of meters, lacking data from longer ranges. This limits models trained on such datasets to handle detection requirements of hundreds or even thousands of meters in real anti-UAV applications. Furthermore, drone swarms represent a significant future threat, yet current datasets cannot support detection research targeting swarm-based threats. Finally, systematically covering drone flight modes (such as hovering, acceleration, cruising, and diving) helps comprehensively capture acoustic feature differences under various operational conditions, enabling refined behavior recognition and situational awareness. Therefore, establishing a large-scale, highly diverse acoustic dataset covering ultra-long-range and swarm scenarios represents a critical bottleneck for advancing algorithms from laboratory research to practical application in this field.

C. Resistance to strong background noise interference

Substantial background noise interference is one of the most prominent challenges in acoustic-based UAV detection. In real-world scenarios, random interference sources are ubiquitous, affecting recorded audio quality to varying degrees. Due to the low signal-to-noise ratio (SNR) of UAV radiation noise received by acoustic arrays, the UAV's rotor noise is often drowned out by background noise, severely limiting detection accuracy. Developing systems that can effectively handle substantial noise interference is crucial for adapting to complex urban environments, making it highly significant. Advances in directional detection, blind source separation techniques, and advanced filtering methods have made significant strides in addressing this issue, and they represent a promising and challenging research direction.

D. Multiple target detection of UAVs

Most current research focuses on detecting one or a small number of UAVs, and representative achievements are still relatively scarce. As UAVs become increasingly cost-effective, detecting UAV swarms will become an unavoidable challenge. In this situation, distinguishing and identifying signals from different UAVs while avoiding signal overlap and false detections presents an important avenue for future research.

E. Precise localization of UAVs

Most research in acoustic-based UAV detection at present is centered on identifying the presence of UAVs or differentiating between categories, with little emphasis on using acoustics for precise UAV localization. By integrating sound source localization

technology, there is potential to create more accurate detection algorithms that can fulfill the growing needs of future detection systems, which is an auspicious research direction.

F. Application of deep reinforcement learning technology

Deep reinforcement learning (DRL),¹⁴⁷ a novel tool, has been widely adopted in fields such as computer vision, healthcare, robotics, and audio signal processing. By integrating reinforcement learning with deep learning, DRL addresses the challenges posed by high computational requirements and the dependency on large datasets for deep learning model training. It has gained substantial attention in the field of audio processing, particularly in solving localization problems. With the growing interest in future audio processing technologies, DRL presents a promising research direction for advancing acoustic-based UAV detection.

VII. CONCLUSION

The acoustic-based UAV detection paradigm has proven to be an economical, passive, and environmentally adaptable solution with fundamental value in addressing the growing security concerns posed by unauthorized drone activities. We systematically reviewed the technical framework, including signal acquisition, pre-processing, feature extraction, identification algorithms, and sound source localization, to clarify the significant achievements and ongoing challenges in this rapidly developing field. A detailed review of the technical workflow reveals critical interdependencies, where high-fidelity signal collection, refined preprocessing to eliminate interference, and subsequent extraction of significant features from both the time and frequency domains are crucial for accurate drone detection. With the continuous maturation and refinement of deep learning technologies, they have demonstrated robust representation learning and generalization capabilities in drone acoustic detection and recognition tasks. Sound source localization represents a key advancement, extending acoustic-based detection from simple presence identification to comprehensive spatial perception. By leveraging array processing and complex sound source localization techniques based on time, angle, and phase differences, precise tracking and threat assessment can be achieved. Although existing technologies have made significant progress, some fundamental challenges remain in terms of noise robustness, long-range detection, and standard datasets. In future research, multimodal data fusion, efficient detection and recognition frameworks, and advanced signal processing techniques are expected to continue advancing detection accuracy, range, multi-target detection, and large-scale deployment. Developing intelligent adaptive systems capable of automatically learning and adapting to environmental changes represents an auspicious direction.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

C.K. and Q.H. contributed equally to this work and share first authorship.

Chengyang Kang: Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Software (equal); Writing – original draft (equal). **Qinyuan Huang:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Fei Sun:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Project administration (equal); Visualization (equal). **Xiuchen Liang:** Conceptualization (equal); Data curation (equal); Investigation (equal); Methodology (equal); Software (equal). **Lan Xu:** Conceptualization (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal).

DATA AVAILABILITY

The data that support the findings of this study are openly available in DroneAudioDataset at <https://github.com/saraalemedi/DroneAudioDataset>.⁴⁰

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